

An Updated Study for the AIAA Aerodynamic Design Optimization Discussion Group Test Case-4

Stephen T. LeDoux ^{*} David P. Young [†] Spencer Fugal [‡] Jonathan Elliot [§]
William P. Huffman [¶] Robin G. Melvin ^{||}

The Boeing Company, Seattle, Washington, 98027. USA.

The fourth model problem from the AIAA Aerodynamic Design Optimization Discussion Group is studied, referred to below as the ADODG CRM wing test case. Two optimization software packages are utilized, TRANAIR design and MDOPT (using both Design of Experiments methods as well as gradient based methods) with OVERFLOW. The resulting designed and associated baseline geometries are cross-analyzed using OVERFLOW and TRANAIR. Previous results using TRANAIR design were run using standard engineering practices and resulted in somewhat non-optimal wings. New results generated using more cycles of design have significantly lower drag and are closer to optimal. In addition, we examine using gradients from an adjoint method in OVERFLOW with the MDOPT tool.

^{*}Associate Technical Fellow, P. O. Box 3707, MC 42-58, AIAA Senior Member.

[†]Technical Fellow, P. O. Box 3707, MC 42-58, AIAA Senior Member.

[‡]Aerodynamics, P. O. Box 3707, MC 42-58.

[§]High Speed Aerodynamics, P. O. Box 3707, MC 42-58

[¶]Geometry and Optimization, P. O. Box 3707, MC 42-58.

^{||}Associate Technical Fellow, P. O. Box 3707, MC 42-58, AIAA Senior Member.

Nomenclature

a	=	Local speed of sound
b	=	Wing Span
CFD	=	Computational Fluid Dynamics
C_D	=	Drag Coefficient = $\frac{Drag}{q_\infty S_{ref}}$
C_L	=	Lift Coefficient = $\frac{Lift}{q_\infty S_{ref}}$
C_M	=	Pitching-Moment Coefficient
C_P	=	Pressure Coefficient = $\frac{P-P_\infty}{q_\infty}$
C_{ref}	=	Wing Reference Chord $\simeq$ MAC
C_f	=	Local Coefficient of Skin Friction
$count$	=	Drag Coefficient Unit = 0.0001
DPW	=	Drag Prediction Workshop
IBL	=	Integral Boundary Layer Method
M	=	Mach Number = $\frac{V}{a}$
P_0	=	Total Pressure
q	=	Dynamic Pressure = $\frac{1}{2}\rho V^2$
$RANS$	=	Reynolds-Averaged Navier-Stokes
Re	=	Reynolds number = $\frac{\rho_\infty V_\infty C_{ref}}{\mu_\infty}$
S_{ref}	=	Reference Area
α	=	Angle-of-Attack
Δ	=	Difference in Quantity (New - Baseline)
η	=	Fraction of Wing Semi-Span
∞	=	Signifies Freestream Conditions

I. Introduction

In our previous work,¹ we presented computational results for the ADODG test cases 1, 2, and 4. We consider cases 1 and 2 to have been disposed of by¹ in which we showed that these single point design problems are ill-posed and tend to produce geometries which for case 1 are singular in inviscid simulations, and for case 2 are singular in IBL and RANS simulations. We also showed how multipoint design for case 2 with a relatively large number of design conditions can result in better behaved designs even though drag increments can still be code and viscous model dependent.

Here we continue the work on case 4, referred to as the ADODG CRM wing. There are several new multipoint design test cases we address, as well as repeating our earlier TRANAIR designs with more design cycles to obtain more optimal wings. We examine the effect of adding thickness modes for the blunt base along with the original camber, thickness, and twist mode shapes and quantify that effect. We also examine the effect of various regularizations and constraints for the problem that can result in more conventional pressure architectures while showing little degradation in drag levels.

Lyu et al² made available the geometry from their single point design for this ADODG CRM case as well. We also compare our wings to this wing and analyze it as well.

We note that these are academic test cases and our results do not reflect formulations that will result in effective wing designs. As demonstrated in,¹ optimization if not properly constrained usually results in defective designs and it is to be stressed that well posed formulations can be very difficult to derive without extensive engineering and mathematical experience.

II. CFD Analysis Methods

In the present study, two well-validated CFD methods are utilized to perform assessments of the aerodynamic designs, namely OVERFLOW and TRANAIR. This section provides a very brief description of each.

OVERFLOW³ is a general-purpose CFD method developed by NASA in the early 1990s. OVERFLOW is capable of solving either the three-dimensional Euler or RANS equations using multiple overset structured grids. It can be applied to complex geometries and has a variety of discretization methods and turbulence modeling options.

TRANAIR⁴⁻⁶ is a general geometry CFD tool that implements a direct coupling between a full potential method which uses a locally refined solution adaptive Cartesian mesh in the field and an integral boundary layer method due to Drela.⁷ TRANAIR uses Newton Krylov methods to achieve reliable convergence over a broad range of flows.⁶ In addition it has a directly coupled aeroelastic capability which, coupled with its rapid cycle time, makes it a viable MDAO tool. In addition, TRANAIR can be extended to many classes of moderately separated flows as discussed in.⁶ Work in underway to include an Euler inviscid solver which in principle enables modeling flows throughout the flight envelop.

III. Aerodynamic Optimization Methods

In the present work, two different approaches are applied to perform the various aerodynamic shape optimizations for the model problems under study. These methods are MDOPT and TRANAIR.

MDOPT⁸ is a Boeing multidisciplinary design optimization framework for general air vehicle design and analysis. The system contains a collection of technology modules for performing optimization studies by combining state of the art numerical optimization schemes with higher-order computational analysis. A variety of multidisciplinary objective and constraint functions are available, including aerodynamic, weight, mission performance, and stability and control characteristics. MDOPT's environment helps manage the tasks of: 1) design-space set-up, 2) establishing a design of experiments, 3) fitting the response surface, 4) navigating the response surface to the optimum state, 5) perturbing field grid points, and 6) enforcing a variety of nonlinear constraints. MDOPT can be exercised in two modes. The first is based on response surfaces.⁹ The second is based on a Boeing developed Sequential Quadratic Programming method using the BFGS Hessian updating method¹⁰ which can be used with finite difference gradients or with the Boeing developed OVERADJ method^{11,12} to supply the gradients. MDOPT is most frequently used with OVERFLOW or TRANAIR as the underlying analysis method.

Many of the results we report were done using OVERADJ. OVERADJ is an "exact reverse" adjoint sensitivity analysis solver based on Chimera overlapping grids, for Euler and turbulent Navier-Stokes physics, and both multigrid and approximate factorization relaxation (using either Pulliam-Chaussee pentadiagonal scheme or the Beam-Warming scheme). It is based on the OVERFLOW-2 flow solver and most of the OVERFLOW options are available for adjoint analysis including Baldwin-Lomax, Spalart-Allmaras and SARC turbulence models, as well as central difference and Roe Upwind schemes, with or without low Mach number preconditioning. Either QCR (the quadratic constitutive relation) or the traditional linear Boussinesq approach can be used to model the turbulent Reynolds stresses. Thin Layer or Full Navier-Stokes terms can be used for viscous terms. Furthermore, exact adjoints can be calculated for flow solutions performed with either fixed freestream angle of attack or fixed lift coefficient. The adjoint solver is based on the discrete adjoint approach and so produces derivatives that are exact regardless of the grid fineness level assuming that both the analysis problem and the adjoint problem are fully converged - unlike the continuous adjoint approach. The approach used results in exact agreement (to within roundoff) at the end of every iteration between the adjoint sensitivity and a sensitivity calculated using a direct method developed in parallel with OVERADJ. Since the direct method uses a solution method identical to the flow solver, both direct and adjoint approaches are guaranteed to produce convergence as long as the flow solver convergence has reached its linear asymptotic state. In practice we find that, although occasionally some work needs to be done to ensure adequate flow solver convergence, this approach gives robust convergence of the adjoint sensitivity even for complex, production quality grids and solution set-ups as shown by its use in numerous design exercises. However, there are many cases where the gridding method is not smooth as the design parameters change, for example if the interblock interpolation stencil changes. This can result in convergence issues for the optimization method even though in many cases, these can be overcome by restarting the BFGS method. Parallelism is achieved by splitting the blocks as in OVERFLOW and distributing the work by assigning each block to different CPUs which communicate using MPI. Open MP was an option available for an earlier version based on OVERFLOW-1 and will be implemented shortly for this OVERFLOW-2 based capability since this mode of parallelism avoids stability issues associated with splitting the blocks.

TRANAIR design¹³ includes the capability of parameterizing the geometry with a variety of mode shape

functions, including camber-and-thickness splines or using surface-curvature polynomials. For camber-and-thickness, any number of control points can be specified. Instead of perturbing the geometry directly, TRANAIR utilizes a transpiration boundary condition to invoke a virtual geometric change to the dividing streamline, which eliminates the grid movement issue that limits many other design methods. On each specified grid of a solution adaptive design run, the flowfield grid is fixed and a sensitivity based subproblem is set up that is solved by NPSOL.¹⁴ This subproblem is consistent to machine precision and thus provides a problem that at least in principle, but usually **not** in practice, NPSOL should be able to fully solve. After each NPSOL solution, one has the option to adapt the grid to the solution. The flowfield solution is then updated on this adapted grid, and another NPSOL subproblem optimization is performed. This process can be repeated until the flowfield solution is sufficiently grid converged. At this stage, the design geometry is relofted using the final transpiration distribution at the boundary. A TRANAIR analysis of this relofted geometry often closely matches that of the virtual geometry. However, in difficult design situations, this may not occur, and multiple design-reloft cycles are required to achieve consistency between TRANAIR solutions on the virtual and real geometries. TRANAIR has a fully coupled aeroelastic capability that also utilizes the transpiration approximation. It is routinely used in this way for loads analysis as well as occasionally for aerostructural optimization.

IV. Case-4 CRM wing optimization test case

Case-4 Model Problem

The ADODG test case 4 is based on the NASA Common Research Model (CRM).¹⁶ Here, the wing of the CRM Wing/Body configuration has been extracted from the theoretical geometry definition and transformed by Osusky¹⁷ as follows. The exposed wing of the CRM WB configuration is translated in the negative span direction to place the root airfoil at the symmetry plane, $y = 0$. The wing is also translated in x and z to place the root leading-edge point at the origin. Finally, this geometry has been scaled down by its mean aerodynamic chord (MAC). We refer to this geometry as the ADODG CRM wing, and in its system, the reference quantities are: $C_{ref} = 1.0$, $S_{ref}/2 = 3.407014$, $X_{ref} = 1.2077$, $Y_{ref} = 0.0$, $Z_{ref} = 0.007669$, with a semispan of $\frac{b}{2} = 3.75820$.

The ADODG test case 4 is broken into several subproblems. For the first one referred to as 4.1 and labeled here as *crm41*, the objective is to minimize the drag of the ADODG CRM wing at the flow condition of $M_\infty = 0.85$, and $Re = 5 \times 10^6$, considering a fully-turbulent flow, and subject to the following constraints.

$$\begin{aligned} C_L &= 0.5 \\ C_M &\geq -0.17 \\ Volume &\geq Volume_{initial} \end{aligned} \tag{1}$$

where, *Volume* refers to the internal volume of the wing. Additionally, the wing thickness at any point cannot decrease by more than 75%.

For most of the new TRANAIR designs this limit was tightened by requiring $C_M \geq -0.16$ instead of -0.17 . This was done to compensate for the potential differences between TRANAIR and RANS/SARC/QCR predicted values for the baseline configuration and give a fairer comparison of the methodologies.

Case-4 MDOPT Single Point Design Results

Three single point designs were done with MDOPT. The mode shape setup had 38 design variables. There were 5 spanwise stations at $\eta = 0.0, 0.25, 0.50, 0.75$, and 1.0 and at each station, there were 4 camber knots located at x_{oc} of $0.2, 0.4, 0.6$, and 0.8 and in addition a leading edge droop mode and a twist mode at each station with the exception of the root section. There was also a single mode for the maximum airfoil thickness value at each station and a single mode controlling the x_{oc} location of this maximum thickness.

For reasons of computer cycle time all these designs were done with a single grid cell spanning the blunt base and closing on its downstream side to an edge. Thus the blunt base was not resolved and the trailing edge thickness was not modified. The OVERFLOW grid for design had a total of 3.2 million grid cells and the wing grid was $289 \times 96 \times 73$. There was an additional wing tip block. The overflow settings for the global and local gradient based methods differed due to convergence limitations. For the global design space

search method the solver setting used in Overflow were the more robust options of the SSOR algorithm coupled with HLLE++ up-winding. These options provided the best convergence for the widely varying geometries required for the initial set of design of experiments sites. The Overflow Adjoint options were limited to the scalar pentadiagonal scheme with central differencing (the current adjoint version does not have SSOR or HLLE++ code differentiated)

The first design was performed with MDOPT in global search mode using Design Explorer (DE) and the SEQOPT optimizer will be referred to as crm41-mdopt38de. This optimization involved computing an initial set of sites from a Design of Experiments (DOE) Orthogonal Array (OA) of 512 runs with sequences of optimization and surrogate model refinement up to a total of 1303 solutions. The convergence for the crm41-mdopt38de case is shown in Figure 1. In the MDOPT OVERFLOW analysis the baseline configuration did not meet the CM constraint of -0.17, and the optimization pushed the first feasible site up to 230 counts of drag (in the plot it is 2.3 as the objective is CD scaled by 100). Improvement in the optimization objective moved below the seed after 40 additional solutions. The optimization was stopped during a poll refinement process before final convergence, as the polling process in the DE mode is highly inefficient with full polls of 38 design variables occurring at each of the iterations with no use of an approximate hessian. In order to refine this optimization further a gradient based finite difference optimization was started from the final DE site 1303 and is labeled as crm41-mdopt38fd. We chose to use a gradient method based on finite differences for this case because the solver settings used for the DE runs are not available with OVERADJ.

The same mode shape setup with 38 degrees of freedom was run with the Boeing developed OVERADJ adjoint OVERFLOW capability to provide gradients to a gradient based optimization method driven by MDOPT. A second OVERADJ/MDOPT design used 143 geometric mode shapes (a subset of the 157 mode shape setup described below in the Tranair setup for crm41-tran157). For the MDOPT case the trailing edge included angle variables were omitted because they were not readily available in MDOPT. This design is referred to as crm41-overadj143. There were 12 spanwise stations and at each station there were 5 camber, 6 thickness. There were 11 twist modes (no root section). Optimization convergence results these single point optimizations are shown in Figures 2 and 3. Neither of these adjoint optimizations fully converged. The objective of the 143 design variable wing is about 1 count better at the design CL of 0.5. The polar data for these wings along with the baseline (using the single grid cell trailing edge closure) is summarized in the idealized profile drag polar plot in Figure 4. Both wings have similar performance at the higher CLs but at lower CLs the 143 design variable wing is better.

It should be pointed out that the OVERADJ results reported here are a first attempt at solving these problems, in what is a relatively new capability. It should be noted that both crm41-overadj38 and crm41-overadj143 should be considered not completely converged though the 38 design variable case is closer. Indeed the 143 design variable case was interrupted by a computer reboot after 51 major iterations - and was not restarted. Other factors that may have reduced the optimization benefit include the use of the single grid cell blunt trailing edge closure model, use of coarse grids, use of different solver settings than those used for ultimate OVERFLOW evaluation (central difference and HLLE++, respectively), and omission of a trailing edge included angle variable. The authors hope to further investigate these limitations in future work.

Case-4 TRANAIR Single Point Design Results

Several designs were performed using TRANAIR single and multi-point optimization. This geometry had a blunt trailing edge which has significant thickness on the outboard wing and was modeled with transpiration and the wake model due to Drela.¹⁸ A standard set of “coarse” camber and thickness mode shapes at 12 spanwise stations was used. These were located at $\eta = 0.0, 0.08, 0.16, 0.25, 0.33, 0.47, 0.62, 0.77, 0.85, 0.92, 0.96$, and 1.0. There were 5 camber, 7 thickness variables including leading edge radius and trailing edge included angle, and 1 twist variable at each of these stations. The internal knots were located at xoc locations of 0.07, 0.18, 0.38, 0.62, and 0.83. The wing root twist variable was constrained to be zero so the root twist remained unchanged. In our previous work,¹ we used standard processes to generate improved wings. Based on previous experience, all TRANAIR designs (except crm41-tran55) were handicapped by using a C_M constraint limit of -0.16 instead of the specified -0.17. Because TRANAIR tends to produce more positive C_M values, this will make the problem more difficult for TRANAIR. We found in our earlier work this penalty was on the order of 0.3 counts. However, for the cases reported here, we found that the differences in C_M between TRANAIR and OVERFLOW/SARC/QCR were not as uniform as we had thought. Some of the TRANAIR designs did come in at $C_M \simeq -0.16$ even in OVERFLOW.

We did repeat our previous TRAN150SC single point design using more grids and design cycles. This

new wing will be referred to as *crm41-tran157*. In Figures 5 and 6 we show the new wing compared to the previously reported *TRAN150Sc* wing. The problem setup was the same. The most notable difference is that the new design has a “beak” on the midboard leading edge. This feature appears to allow a lower pressure on the lower surface than a more rounded geometry. The result was an on design drag level that was 1.7 counts lower as measured in *TRANAIR*. We also added a design with trailing edge thickness degrees of freedom at each of the 12 span station for this case. The designed wing will be referred to as *crm41-tran181*. The result was a further 1.7 count drag reduction as measured in *TRANAIR*. The trailing edge thickness became larger inboard since it increases wing volume and decreased outboard to the lower bound. This resulted in considerably thinner airfoil sections overall outboard and a thicker plane of symmetry airfoil. *OVERFLOW* showed that the new *crm41-tran157* wing had 0.1 counts more drag than *TRAN150SC* at the design point, while *crm41-tran181* had 2.5 counts less drag. Thus, the increment for the trailing edge thickness modes would appear to be 2.6 counts. These results are not very consistent with the *TRANAIR* analysis results. In Figure 7 we compare the geometries of *crm41-tran157* and *crm41-tran181*. Figure 8 shows the C_p distributions.

An even courser set of modes was used as well with only 6 span stations and at each station, 3 camber modes, 5 thickness and 1 twist. There were a total of 55 design degrees of freedom and this is referred to as *crm41-tran55*. This case was run with the original $C_M \geq -0.17$ constraint.

A denser set of 283 mode shapes was also used. The 12 span stations were the same ones enumerated above, but there were 9 camber and 11 thickness and 1 twist at each of the stations. The chordwise locations of the interior knots were at x_{oc} values of 0.01, 0.03, 0.06, 0.12, 0.20, 0.38, 0.62, 0.83, and 0.92. The wing root twist is constrained to be zero. The new multipoint designs all used this dense set of mode shapes. This setup also included a trailing edge curvature constraint to limit aft loading.

The new results were run with an enhanced number of design cycles. Each design cycle optimized on a series of nine solution adaptive grids (previous results had used four). The first four grids were coarser grids and the final five were at the same fine level of grid resolution. Three cycles of design, each followed by a reloff of the geometry were used. The result was that reductions in drag were achieved that in some cases appear significant, leading to the conclusion that our previously reported results were in some cases rather far from optimal. In general the new designs were thinner outboard and thicker at the root, in addition to having a tendency to generate a “beak” at the leading edge. The leading edge pressure rises were also sharper. All of the new designs used a trailing edge curvature constraint to control aft loading. This was done because *TRANAIR* tends to like very aggressive trailing edge pressure gradients that may not be physically realizable.

We did several single point designs with the 283 mode shape setup. The first, *crm41-tran283* included the spanwise and chordwise constraints used for *TRAN150SC*. Another design additionally imposed maximum Mach number constraints that varied spanwise based on the baseline maximum Mach numbers and also convexity constraints to prevent beaking near the leading edge and on the upper surface near the trailing edge. This is referred to as *crm41-tran283-convex*.

We felt these new designs were converged using details of the optimization subproblem on each grid even though *TRANAIR* does not produce a rigorous estimate of the reduced gradient as part of a standard run. In all cases, the drag reduction on the final cycle of nine design grids was less than a single count. All constraints were met as discretely formulated, including the wing volume constraint. These computations revealed that the *TRANAIR* designed wings reported previously¹ were rather far from optimal, by roughly 1-3 drag counts depending on which code they are measured in.

Case-4 Multi-point MDOPT Design Results

Optimizations for the multipoint *crm42* case were performed in MDOPT using the global DE and *OVERADJ* methods. The same geometric setups up were used as discussed for the single point designs. These designs are referred to as *crm42-mdopt38de*, *crm42-overadj38*, and *crm42-overadj143*. For the global based design method the results of the refined optimization are shown in the scatter plot in Figure 9. Here the drag at the first two design points are plotted with the total objective on the color scale. This optimization used the original 1303 single point sites as the starting database for the surrogate modeling process and then ran an additional 1529 sites to complete the multipoint MDOPT/DE optimization. With a total of 2832 runs completed this design seems to be fairly expensive and the value of the DE method seems limited. However the real value of the DE global based optimization is the ability to use the refined optimization design space to perform trades on design variables, multipoint weighting changes, Pareto front refinement and complex

constraint trades. For instance we were able to start the multipoint case using the single point database.

OVERADJ optimization results for the multi-point optimization are shown in Figures 10 and 11. The first figure shows the drag at the primary condition only. The second shows the lift at all three design conditions. During the 38 design variable optimization a discontinuity in the wing volume calculator was found and subsequently fixed for the 143 design variable optimization. The discontinuity led to the strange convergence behavior seen the 38 design variables optimization. The ideal profile drag polar for these designs is shown in 12 and the lift curve is shown in Figure 13, both using again the single grid cell trailing edge closure. Including more design variables improves the higher CL design point. Geometry for the wings for the different optimization cases with different design variable setups is shown in Figures 14 15. In general the designs have similar features but differ in detail. In Figure 16 we show the drag polars as analyzed using the single cell trailing edge closure grid used for the design compared to the two mdopt38 single point designs.

Case-4 Multi-point TRANAIR Design Results

For the 157 mode shape setup, in our previous work, a multi-point optimization was performed with 5 design points, i.e., the first 5 conditions listed in Table 1. This wing was referred to as TRAN150MP5.

A second multi-point design was performed in our earlier work and was called TRAN283MP7. The 7 flow conditions used in the objective are shown in Table 1. There were more flow conditions because there were more camber modes. It used the same 283 mode shape setup as our new multipoint designs and described above. To explore the relationship between TRANAIR and RANS modeling, we also limited aft camber to control aft loading. We also introduced modes that allowed the trailing edge bluntness to change as a function of span.

We repeated the TRAN283MP7 multipoint design from our previous work and refer to this new wing as special7-tran283. As measured in TRANAIR, the C_D level was 1.4 counts lower than the previous design at the primary flight condition and the geometry was more beaked near the leading edge. There were several new multi-point designs specified by the ADODG that we ran. The first involved C_L 's of 0.45, 0.50, and 0.55 all at $M_\infty = 0.85$. This design is referred to as crm42-tran283. A second involved $M_\infty = 0.84, 0.85$, and 0., 86, all at $C_L = 0.50$. This is referred to as crm43-tran283. There was also an optional 9 point design referred to as crm46-tran283. The drag increments vs. the baseline are reported in the table below. These designs also showed significant increases in curvature near the leading edge, sharper leading edge pressure expansions, and beaks on the lower surface.

Mach number	Cl	Weight
0.85	0.50	1.0
0.86	0.50	1.0
0.80	0.50	1.0
0.85	0.55	1.0
0.85	0.45	1.0
0.85	0.52	1.0
0.85	0.40	1.0

Table 1. Flight Conditions for the TRANAIR Multi-point Wing Designs.

Case-4 Other Results and Observations

Additionally, we analyzed the single point designed wing of Lyu et. al.² and compared some of the TRANAIR designed wings to it. Our results confirm that this wing is indeed well designed, even though TRANAIR showed the TRANAIR designs to be better (an expected result). OVERFLOW/SARC/QCR showed it to be better than the TRANAIR and the MDOPT designed wings. We hypothesized that one reason for this may be the much larger design space employed by Lyu et al. Using TRANAIR, we ran some single point designs with many more mode shapes, one with 415 and another with 739, to try to confirm this hypothesis. Both of these designs used 18 spanwise stations roughly equally spaced from root to tip. The 415 mode shape case used the same sectional modes as crm41-tran283, i.e., 9 camber, 11 thickness, and 1 twist mode at each station. The 739 mode case used 18 camber, 20 thickness, and 1 twist mode at each station. The knots

for camber and thickness were more closely spaced near the leading edge but in no location more than 6% of chord apart. Our first crm41-tran739 run had no smoothness constraints except the very lax curvature bounds used with coarser mode sets and found a design with rather severe wiggles on the lower surface. Convergence stagnated at this “solution”. In Figure 17 we show the mach distribution for this wing. We also reran this case with our standard smoothness encouragement penalty function but final results were not available for this paper. The smoothness penalty was also required to obtain reasonable convergence for the crm42-tran283 design for example.

Using the SST turbulence model, a different set of results was obtained with OVERFLOW and the same grids as used with the SARC/QCR model. The SST model gives results that appear to agree better with TRANAIR than SARC/QCR or other versions of SA examined in our earlier work. This result should provide strong evidence that viscous modeling details will have a strong influence not just on absolute drag levels, but even on drag increments, polar shapes, and pitching moment values. However, the residual convergence of the SST model in OVERFLOW was not as robust as the SARC/QCR model.

We investigated the effect of using the single grid cell trailing edge closure in the MDOPT designs. In Figure 18 we show OVERFLOW polars for the baseline wing and crm41-mdopt38de run with this single cell closure. Also shown are results from our standard gridded up trailing edge OVERFLOW analysis using a cap grid. There is a polar rotation for the baseline wing and a change in shape for crm41-mdopt38de. Figure 19 shows the pitching moment values and shows that there is a shift in C_M on the order of 0.01. This shift in effect handicapped the MDOPT designs. However, interestingly, as measured in the single cell closure grid method used in the optimization, the designed wing was only about 7.5 counts better than the baseline at the design point, whereas using the different analysis method with the gridded up trailing edge, the increment was a larger 9.1 counts.

We did not do a detailed comparison of computer wall clock or CPU times. MDOPT using design of experiments scales poorly with number of design variables. For the MDOPT/OVERADJ and TRANAIR the scaling is better, but still far from optimal. For MDAO applications, for example wing sizing, cycle time is critical and in fact adjusted doublet lattice is the method used exclusively in production. Generally, the TRANAIR designs took on the order of 36 core hours for each cycle or a total of 108 core hours for an entire design, about triple the cycle time for the designs reported in our previous work.¹

In Figure 20 we show OVERFLOW results obtained for some of our single point designed wings using the classic version of the SA turbulence model¹⁹ with the trip terms commonly referred to as the Ia version and the Rotational Correction for some of the single point designed wings. This will be referred to as SARC. This model was run with the Quadratic Constitutive Relation correction (QCR). We believe these results to be nearly grid asymptotic and converged by usual engineering standards, even though not by mathematically rigorous standards as employed in TRANAIR. Figure 21 shows TRANAIR polars for these wings. Both crm41-tran157 and crm41-overadj143 used essentially the same mode shapes and their polars are roughly comparable in OVERFLOW, but not in TRANAIR where the TRANAIR wing is substantially better, an expected result. In Figure 22 we show the pitching moment data from TRANAIR. Figure 23 shows the TRANAIR computed drag rise curves for some of these wings.

In Figures 24, 25, and 26, we show TRANAIR, OVERFLOW/SARC/QCR, and OVERFLOW/SSST polars for some of the crm42 multipoint test case wings.

In Figures 27 and 28, we compare the crm41-tran157 design with the crm41-overadj143 design. The latter is strikingly thicker outboard and has increased aft loading over the baseline, even though not as much as crm41-tran157. This indicates that the redistribution of volume inboard is a subtle effect as measured by either OVERFLOW or TRANAIR and will only take place in the final stage of convergence. The crm41-overadj143 design stopped prematurely before full convergence due to a hardware problem.

In Figures 29 and 30, we show the polars from OVERFLOW with the fully gridded trailing edge and TRANAIR for crm41-tran157, crm41-tran181, and crm41-tran283. While the results in TRANAIR are consistent and very similar, in OVERFLOW, the polars have the same shape but have seemingly arbitrary shifts from each other.

In Figure 31, we show the polars from OVERFLOW for crm41-mdopt38de, crm41-mdopt38fd, and crm41-overadj38. These polars seem to show that the overadj wing is the most optimal, followed by the mdoptfd wing, with the mdoptde being the furthest from optimal.

In an attempt to quantify the turbulence modeling uncertainty for these wings, we also ran OVERFLOW with the SST turbulence model using the same grids as SARC/QCR for the baseline wing and crv41-tran181. Figure 32 shows the polars. The increments are very dependent on the details on viscous modeling. Drela’s

IBL has the thickest boundary layers, followed by SST, and then SA in all its incarnations. The polars also do have somewhat different shapes as well. This is perhaps a surprising result to some, but quite consistent with our experience.

While drag polars are a better measure of design effectiveness, we summarize in Table 2 the drag increments for some of the single point designs considered in various analysis codes. Table 3 contains data for the multi-point designs. We caution that these increments are not robust to changes in turbulence modeling, gridding, or other details of CFD codes and methods.

Method	OVERFLOWSAQCR	OVERFLOWSST	TRANAIR
crm41-tran55			20.9
crm41-tran157	12.9		23.1
crm41-tran181	15.3	20.2	24.8
crm41-tran283	13.9		24.8
crm41-tran283-convex			24.0
crm41-tran415			25.6
crm41-mdopt38de	9.1		10.4
crm41-overadj38	12.1		13.6
crm41-overadj143	12.6		13.4
Lyu-et al	17.0		20.6

Table 2. C_D Increments in Counts for Various Methods for the Single Point Designed Wings.

Method	OVERFLOWSAQCR	OVERFLOWSST	TRANAIR
crm42-overadj38	12.4	16.6	16.0
crm42-tran283	13.5	18.3	24.4
crm43-tran283	13.2		24.8
crm46-tran283			22.4
special7-tran283			24.3

Table 3. C_D Increments in Counts for Various Methods for the Multi-Point Designed Wings.

We make a few observations about these results, particularly the drag polars.

1. There is reasonable agreement between polar shapes as predicted by OVERFLOW/SARC/QCR and TRANAIR as shown in Figures 20 and 21. There are different shifts however for different wings, depending on the design method used. TRANAIR's wings are generally better at higher C_l values. Overall, TRANAIR shows substantially larger increments at the design point for its designs than OVERFLOW/SARC/QCR, an expected result. The increment for OVERFLOW/SST as shown in Figure 32, lies nearly halfway between the other two. The polar shapes are also somewhat different. This is perhaps partially explained by the fact that SST in general tends to give thicker boundary layers than SA. For wings designed with other methods, TRANAIR analysis typically showed 3-5 counts of wave drag, not an unexpected result since Drela's IBL boundary layer is typically thicker than RANS boundary layer approximations. Of course, this "extra" drag is absent in TRANAIR analyses of the wings designed in TRANAIR. We emphasize again that wave drag is simply not robust to changes in turbulence modeling. What this shows is simply that optimization takes advantage of "features" of CFD methods that may not be present in other methods. One must conclude that even the drag increments for these wings can be strongly dependent on the viscous model used.
2. As shown in¹ it appears that SARC/QCR agrees better with TRANAIR's predictions than other versions of the SA model, while the SST model is still closer to TRANAIR results as shown in Figure 32.
3. As observed in our previous work, optimization (especially when large changes are allowed and there are few constraints) will exploit "features" of each viscous model being used. It is also true that

these “features” are things that will vary significantly among models. Perhaps this should introduce a cautionary note that the weaknesses of CFD analysis methods will be exploited by effective optimization methods. Being more optimal is thus liable to result in poorer agreement among models, calling into question the naive view that optimization must be a good thing.

4. TRANAIR designs seem to be better at higher C_L ’s in all analysis codes, perhaps due to aft loading. The TRANAIR designs also seem to have delayed pitch up, a desirable characteristic for these swept transonic wings.
5. The MDOPT designed wings have strikingly higher profile drag in TRANAIR than the TRANAIR designed wings (for the single point designs 8-10 counts higher), illustrating the differences in the boundary layer models. One factor in this is that Drela’s integral boundary layer method, for a given inviscid pressure profile, has a thicker boundary layer with more profile drag than the SA model.
6. As also observed in,² there seems to be a weak dependence of the designed drag levels on the number of design variables. There is hardly any difference in drag between the crm41-overadj38 and crm41-overadj143 in both OVERFLOW and TRANAIR or between crm41-tran181 and crm41-tran283 with crm41-tran55 being only a little higher in drag as measured in TRANAIR. crm41-tran415 is only about 0.8 counts better than crm41-tran181 as measured in TRANAIR. Final results are not available for crm41-tran736 at this time.
7. While the TRANAIR wings show consistent drag levels when analyzed in TRANAIR, i.e., more degrees of freedom result in lower drag, this is not the case when these wings are analyzed in OVERFLOW/SARC/QCR. We believe that because of the differences in viscous models which are not small, how well these wings do in OVERFLOW/SARC/QCR is dependent to some extent on effects that are artifacts of the viscous methods involved. The same is true to a lesser degree of the overadj wings when analyzed in TRANAIR. TRANAIR wings had sharp pressure rises at the leading edge and a lot of aft loading, thus accentuating the viscous model differences.

Case-4 Summary Conclusions

1. Small differences in methodology or mode shape definitions can produce significant differences in geometry for this problem.
2. Viscous modeling details can make a big difference in the results for singlepoint and to a lesser extent for multipoint designed wings.
3. More design degrees of freedom for a single point design usually produce a less favorable drag rise curve and produce only a relatively small improvement in drag levels.
4. Multi-point optimization produces better behaved designs with more design points resulting in more favorable drag polars.
5. Differences in values of the C_M constraint have a minor effect on the designed drag levels. This effect seems to be less than half a count. It appears that TRANAIR often, but not always, produces more positive C_M values than RANS/SARA/QCR. C_M values generally involve numerical cancellation and thus are particularly sensitive to grid resolution and algorithm details. The TRANAIR designed wings show a delayed pitch break as measured in TRANAIR.

V. Conclusions

We have further refined our previous study of the ADODG CRM wing test cases, including single and multipoint design cases. Determination of a well posed formulation in each new engineering situation is an unsolved problem and is currently reliant on expert judgment. This is a largely unexplored frontier in CFD and MDAO that would seem to be a significant bottleneck for any airframe optimization process involving advanced CFD, at least given the current state of CFD solvers and theory. Of course, in actual aircraft design, the number of flight conditions to consider is quite large and computationally impossible with high

fidelity CFD codes alone. Thus engineering judgment is required to guide the problem formulation which can be quite complex and invariably involves lower fidelity tools such as adjusted doublet lattice. Thus, mixed fidelity approaches would seem to be required.

However, the academic test cases studied here can offer some insight into the relative performance of various optimization methodologies and offer clues to where further work is needed.

Even a relatively superficial study such as contained herein reveals that the uncertainty in CFD simulations is larger than commonly thought. It is dangerous to generalize from seemingly good agreement in a few selected test cases, especially when the data is included in the uncertainty range of the CFD methods, allowing unconscious selection or positive results bias. A growing recognition of these biases and problems would seem to require more scientifically rigorous validation studies than the one undertaken here.

VI. References

References

- ¹LeDoux, S. T., Vassberg, J. C., Young, D. P., Fugal, S., Kamenetskiy, D., Melvin, R. G., Huffman, W. P., and Smith, M. F., "A Study Based on the AIAA Aerodynamic Design Optimization Discussion Group Test Cases", *AIAA Journal*, (2014), to appear.
- ²Lyu, Z., Kenway, G. K. W. and Martins, J. R. R. A., "Rans-based Aerodynamic Shape Optimization Investigations of the Common Research Model Wing," AIAA Paper 2014-0567, Jan. 2014.
- ³Nichols, R. H. and Buning, P. G., OVERFLOW User's Manual, Version 2.1t. Technical report, U. of Alabama and NASA, August 2008.
- ⁴Melvin, R. G., Young, D. P., Huffman, W. P., Johnson, F. T., Hilmes, C. L., and Bieterman, M. B., "Recent Progress in Aerodynamic Design Optimization," *International Journal for Numerical Methods in Fluids*, Vol. 30, 1999, pp. 205–216.
- ⁵Samant, S. S., Bussioletti, J. E., Johnson, F. T., Burkhart, R. H., Everson, B. L., Melvin, R. G., and Young, D. P., "TRANAIR: A Computer Code for Transonic Analysis of Arbitrary Configurations," AIAA Paper 87-0034, AIAA Aerospace Sciences Meeting and Exhibit, Reno, NV, January 1987.
- ⁶D. P. Young, R. G. Melvin, W. P. Huffman, E. Arian, M. Hong, and M. Drela, "Implementation of a Separated Flow Capability in TRANAIR", *AIAA Journal*, Vol. 52, #8, 2014, pp. 1699–1716.
- ⁷Drela, M., "Two-Dimensional Transonic Aerodynamic Design and Analysis Using The Euler Equations," PhD Dissertation, MIT, 1985.
- ⁸LeDoux, S. T., Herling, W. W., Fatta, J., and Ratcliff, R. R., "Multidisciplinary Design Optimization System Using Higher Order Analysis Codes," AIAA Paper 2004-4567, 10th AIAA/ISSMO Multidisciplinary Analysis and Optimization Conference, 30 August - 1 September, 2004, Albany, New York.
- ⁹A. Booker, "Well-Conditioned Kriging Models for Computer Output", M&CT-TECH-00-002, 2000.
- ¹⁰Betts, J. T. and Frank, P. D., "A Sparse Nonlinear Optimization Algorithm," *Journal of Optimization Theory and Applications*, Vol. 82, 1999, pp. 519–554.
- ¹¹Elliot, J., "Discrete Adjoint Analysis and Optimization with Overset Grid Modelling of the Compressible, High-Re Navier-Stokes Equation," presented at the 6th Overset Grid and Solution Technology Symposium, Fort Walton Beach, FL, Oct, 2002.
- ¹²Elliot, J., and Herling, W., "A Chimera Approach to Aerodynamic Shape Optimization for the Compressible, High-Re Navier-Stokes Equations," AIAA Paper, 2000-4729, Sept. 2000.
- ¹³Young, D. P., Huffman, W. P., Melvin, R. G., Hilmes, C. L., and Johnson, F. T., "Nonlinear Elimination in Aerodynamic Analysis and Design Optimization," in *Large-Scale PDE-Constrained Optimization*, Springer Verlag Lecture Notes in Computational Science and Engineering, edited by Biegler, L. T., Ghattas, O., Heinkenschloss, M., and van Bloemen Waanders, B., Springer, 2003, pp. 17–44.
- ¹⁴Gill, P. E., Murray, W., Saunders, M. A., and Wright, M. A., *User's Guide for NPSOL (Version 4.0): A FORTRAN Package Nonlinear Programming*. Stanford University Technical Report SOL86-2, Department of Operations Research, 1986.
- ¹⁵Drela, M., "Pros and Cons of Airfoil Optimization," *Frontiers of Computational Fluid Dynamics 1998*, edited by Caughey, D. A. and Hafez, M. M., World Scientific, pp. 363-380.
- ¹⁶Vassberg, J. C., DeHaan, M. A., S. M. Rivers, S. M., and R. A. Wahls, R. A., "Development of a Common Research Model for applied CFD validation studies," AIAA Paper 2008-6919, 26th AIAA Applied Aerodynamics Conference, Honolulu, HI, August 2008.
- ¹⁷Osusky, L. M. and Zingg, D. W., "Lift-constrained drag minimization of a wing allowing section and twist variation with flow governed by the Reynolds-Averaged Navier-Stokes equations," ADO DG Sharepoint file: crm_study2July2013.pdf, July 2013.
- ¹⁸Drela, M., "Integral Boundary Layer Formulation for Blunt Trailing Edges," AIAA Paper 89-2200, 1989.
- ¹⁹Spalart, P. R. and Allmaras, S. R., "A One-Equation Turbulence Model for Aerodynamic Flows," *Recherche Aerospatiale*, No. 1, 1994, pp. 5-21.

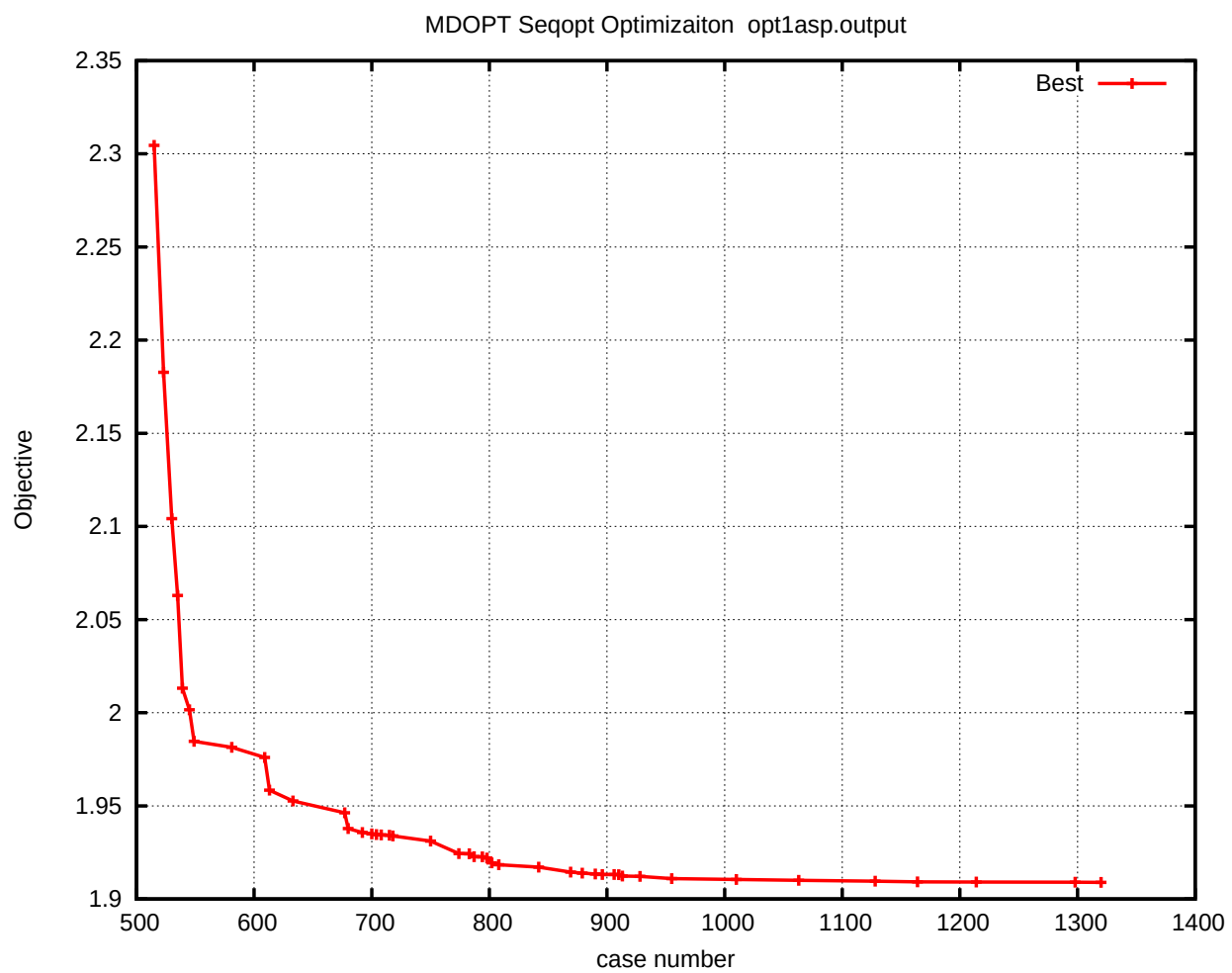


Figure 1. Convergence of the Drag for the crm41-mdopt38de Design.

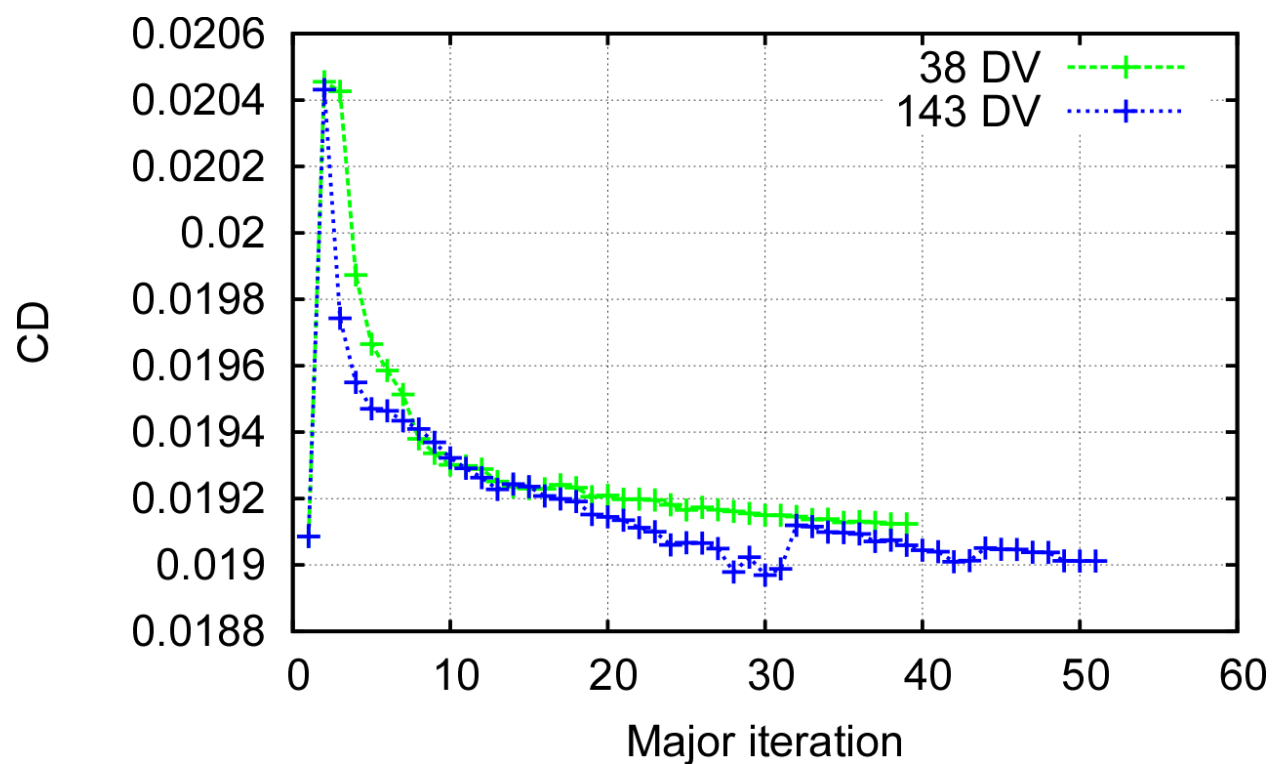


Figure 2. Convergence of the Drag for the crm41-overadj38 and crm41-overadj143 Designs.

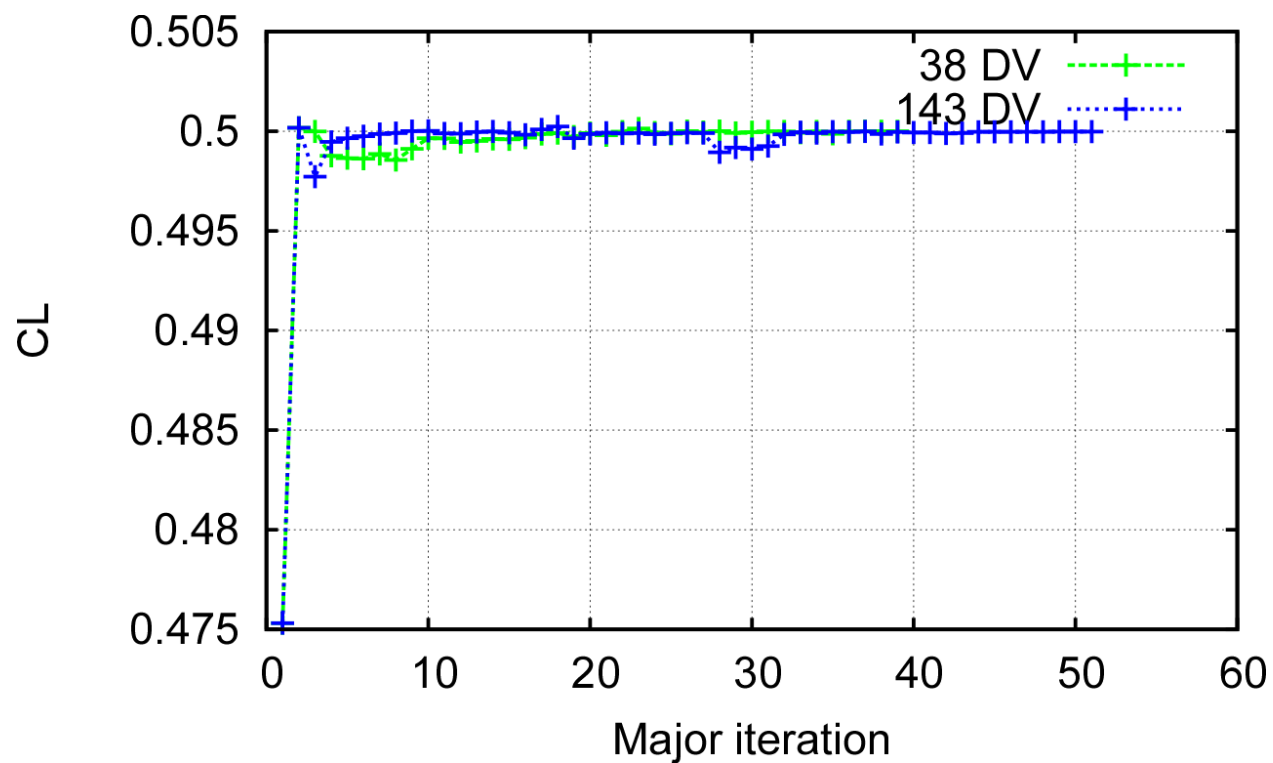
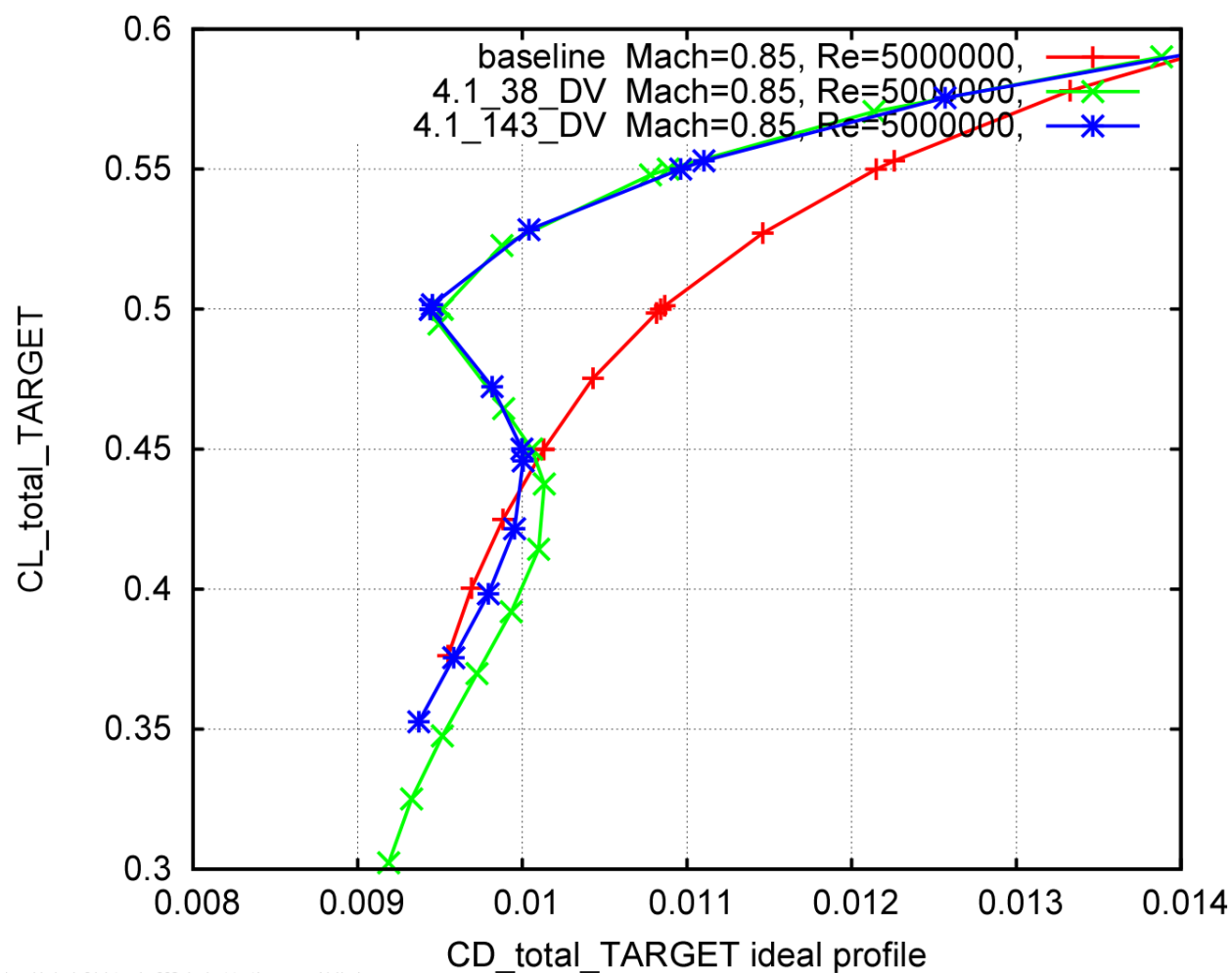


Figure 3. Convergence of the Lift for the crm41-overadj38 and crm41-overadj143 Designs.



hendrix-login01:/ptmp/gy662c/mdopt-test/crm-case41d/subcases

Figure 4. Drag Polars for the OVERADJ crm41-overadj38 and crm41-overadj143 Designs Compared to the Baseline Wing With the Single Cell Trailing Edge Gridding.

Z

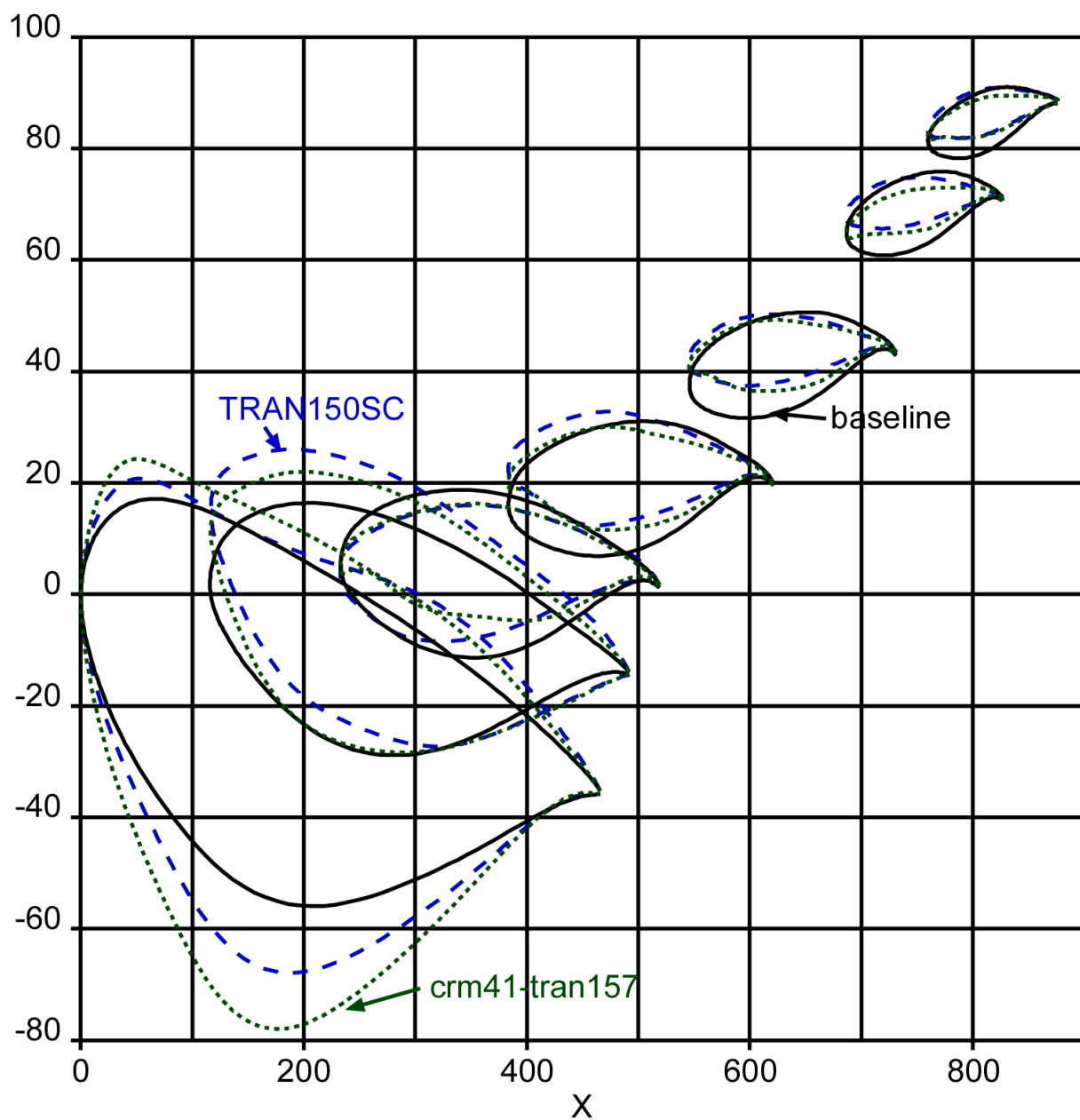


Figure 5. Wing Geometries for TRAN150SC vs. crm41-tran157 vs. the ADODG CRM Baseline.

Cp

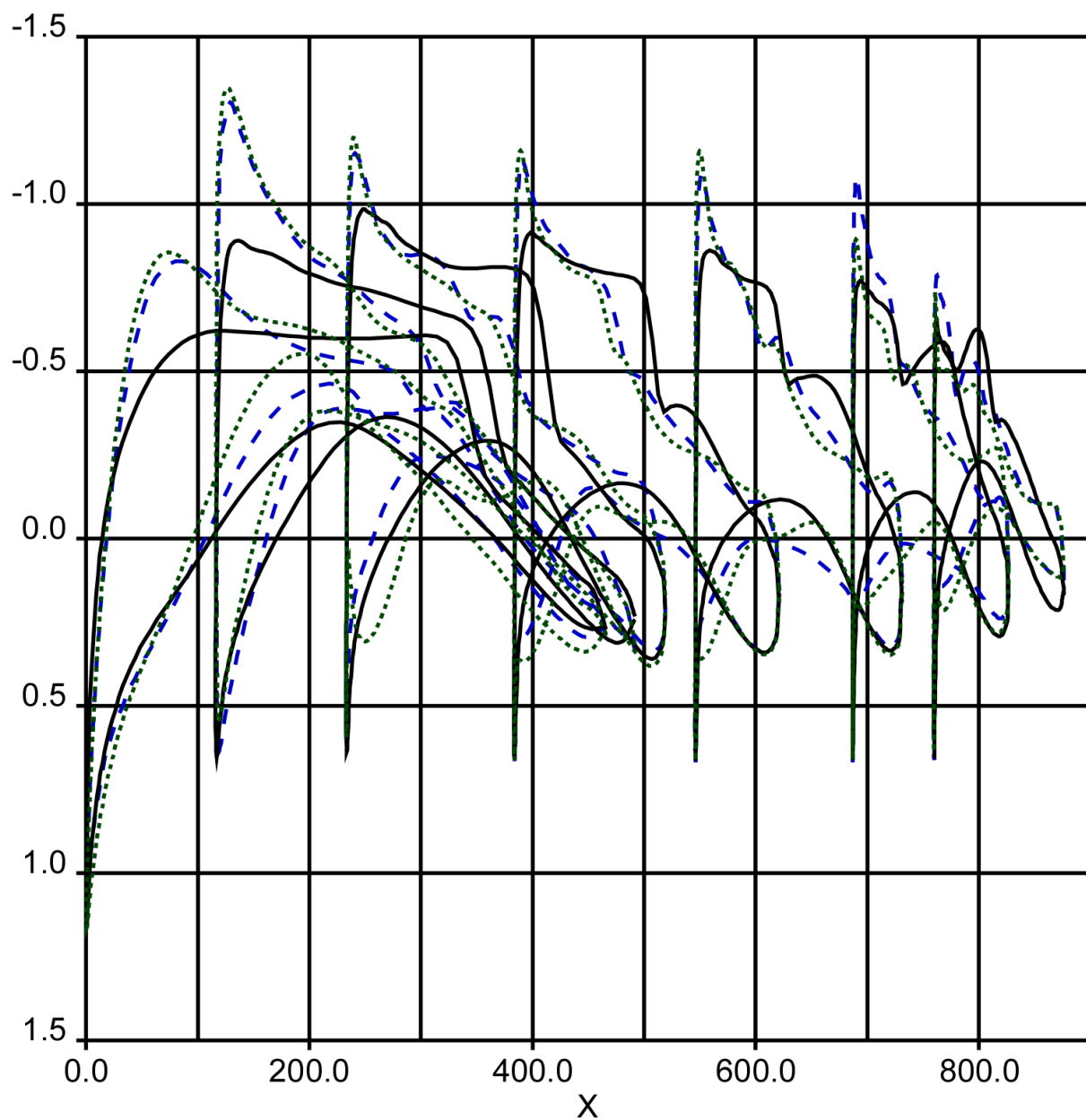


Figure 6. C_p Distributions for TRAN150SC vs. crm41-tran157 vs. the ADODG CRM Baseline.

Z

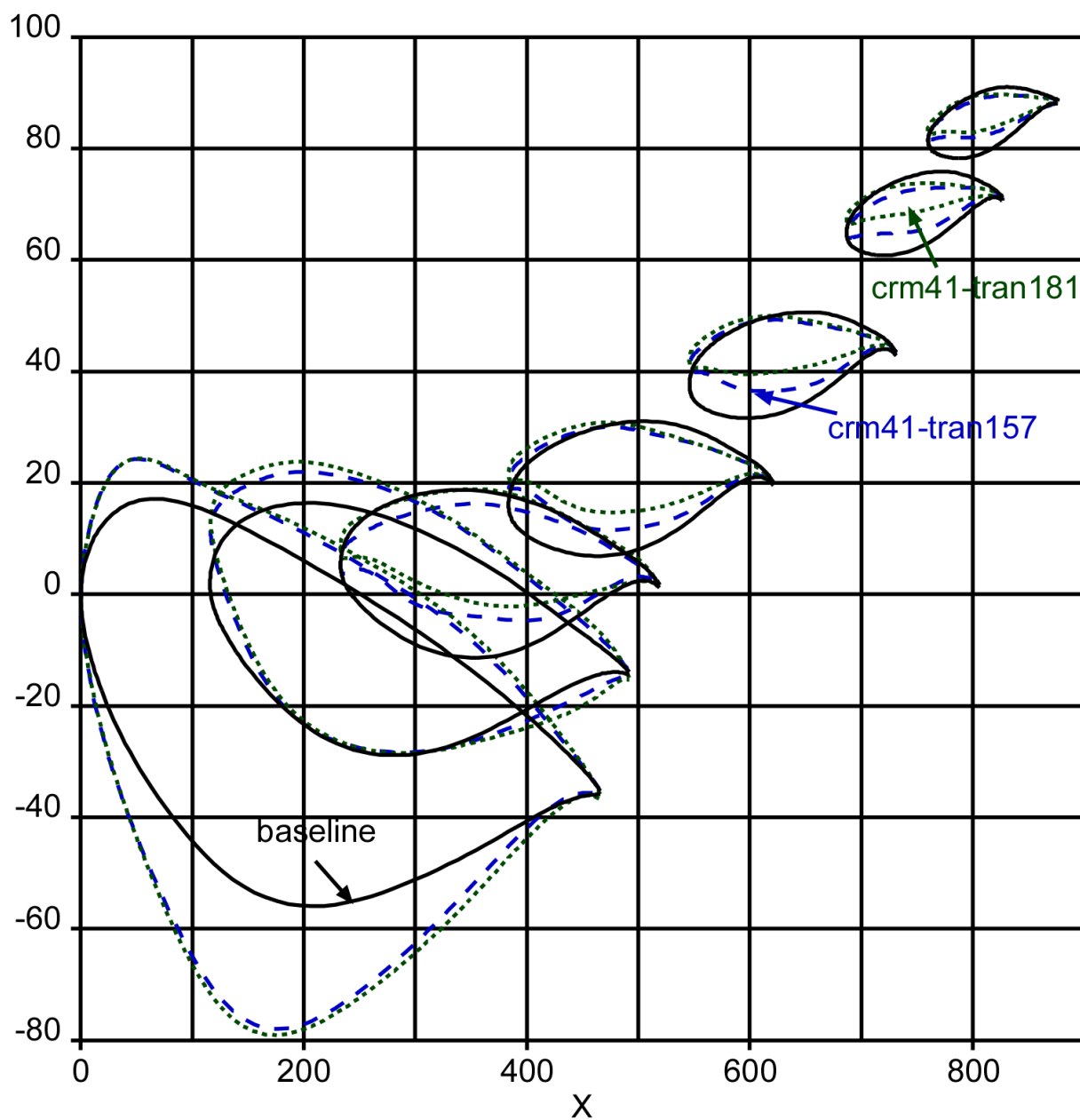


Figure 7. Wing Geometries for crm41-tran157 vs. crm41-tran181 vs. the ADODG CRM Baseline.

Cp

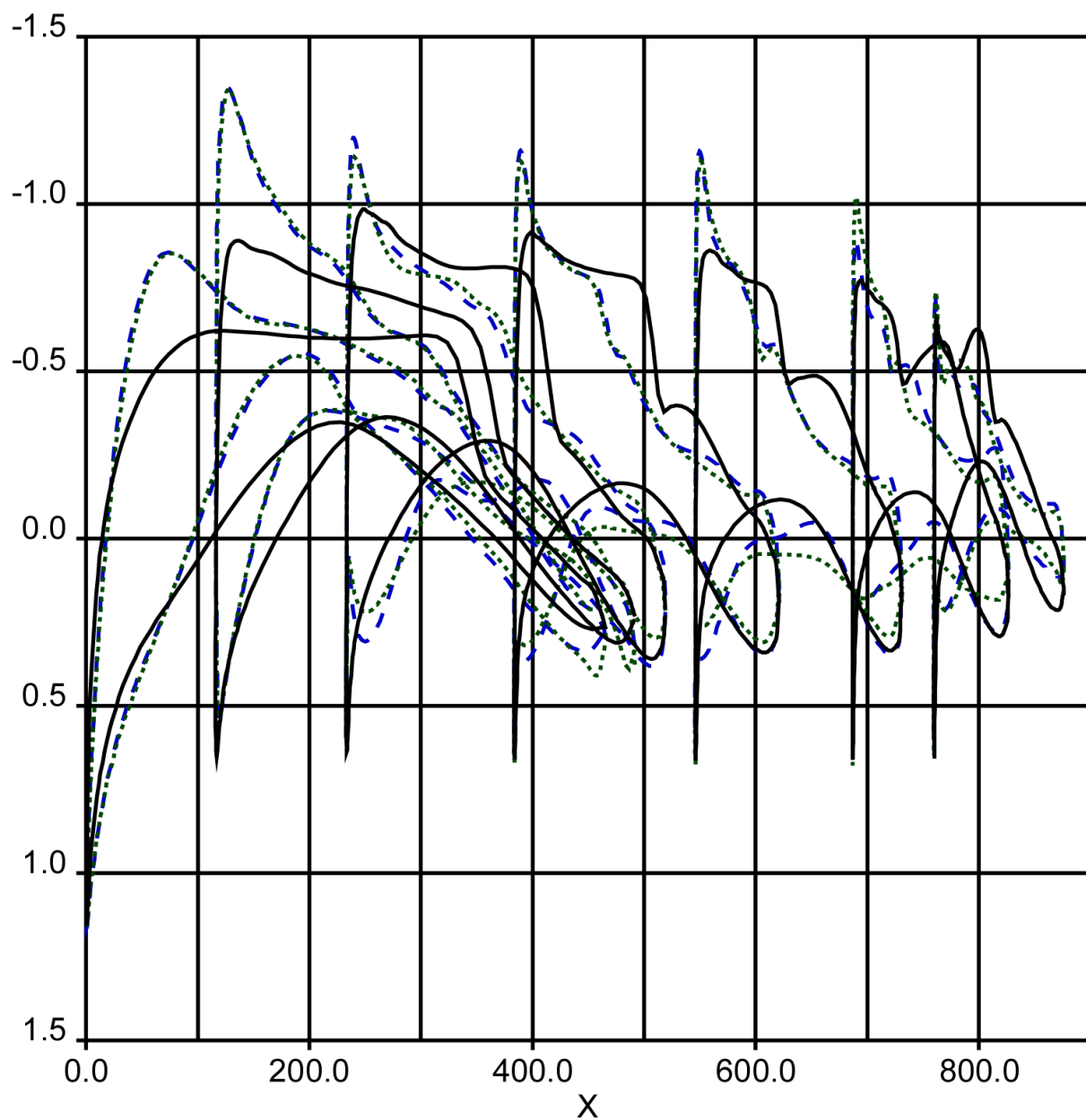


Figure 8. C_p Distributions for crm41-tran157 vs. crm41-tran181 vs. the ADODG CRM Baseline.

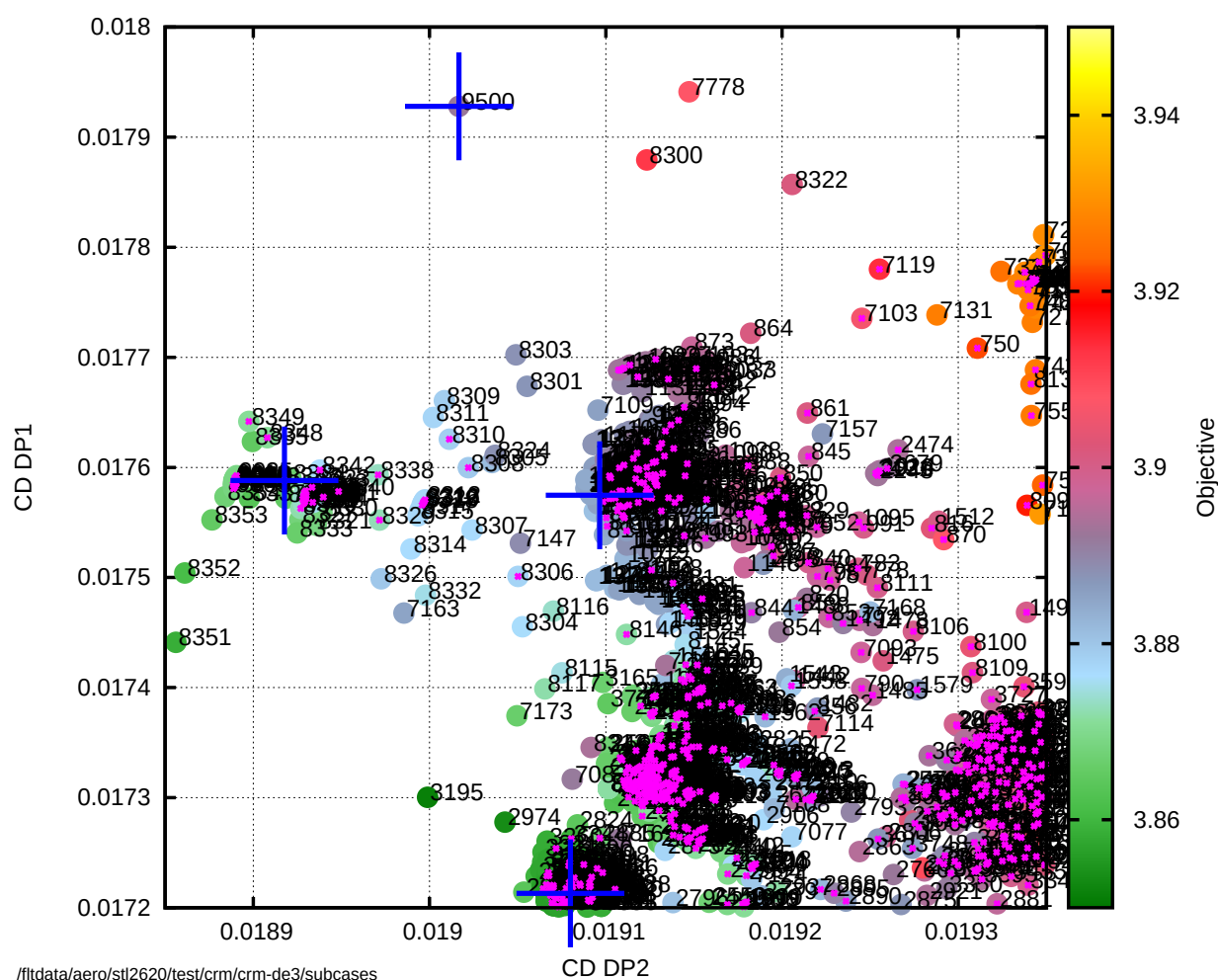


Figure 9. Scatter Plot of Designs examined by MDOPT for crm41-mdopt38de compared to crm41-mdopt38fd and crm42-mdopt38de.

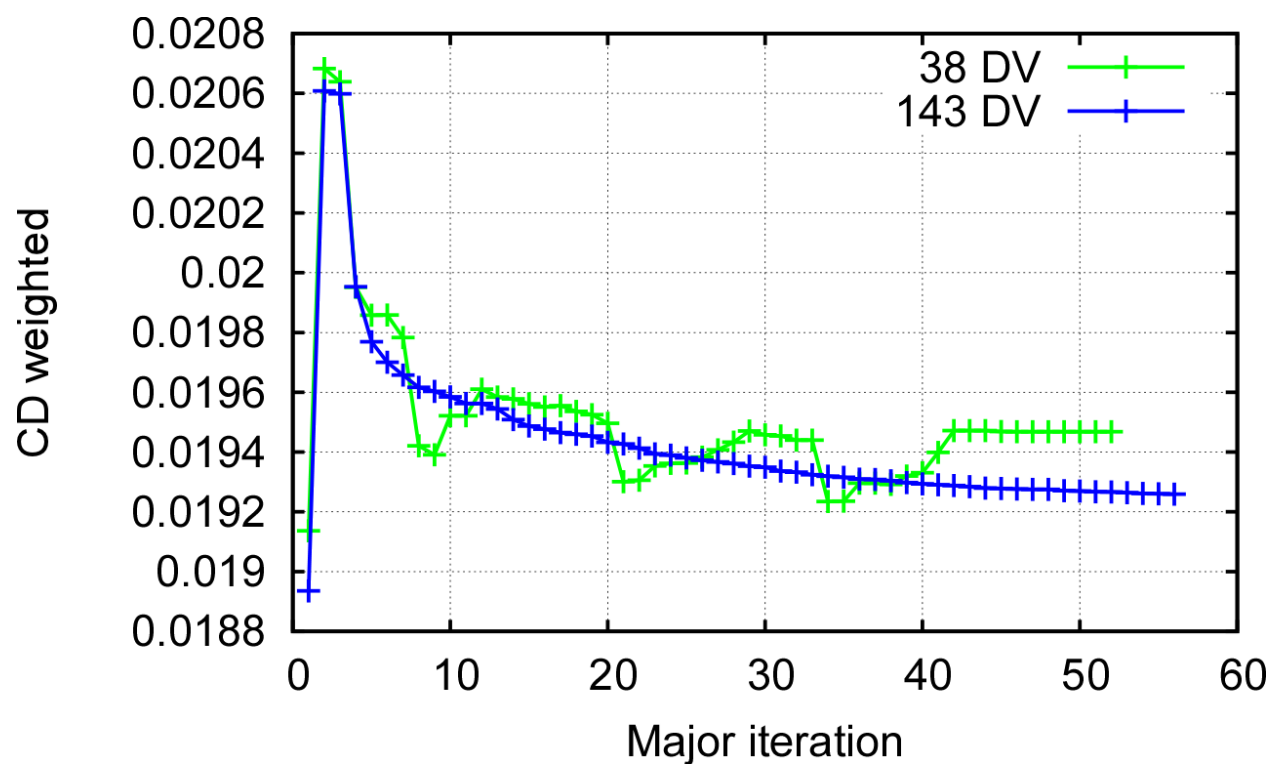


Figure 10. Convergence of the Drag for crm42-overadj38 and crm42-overadj143 Multipoint Designs.

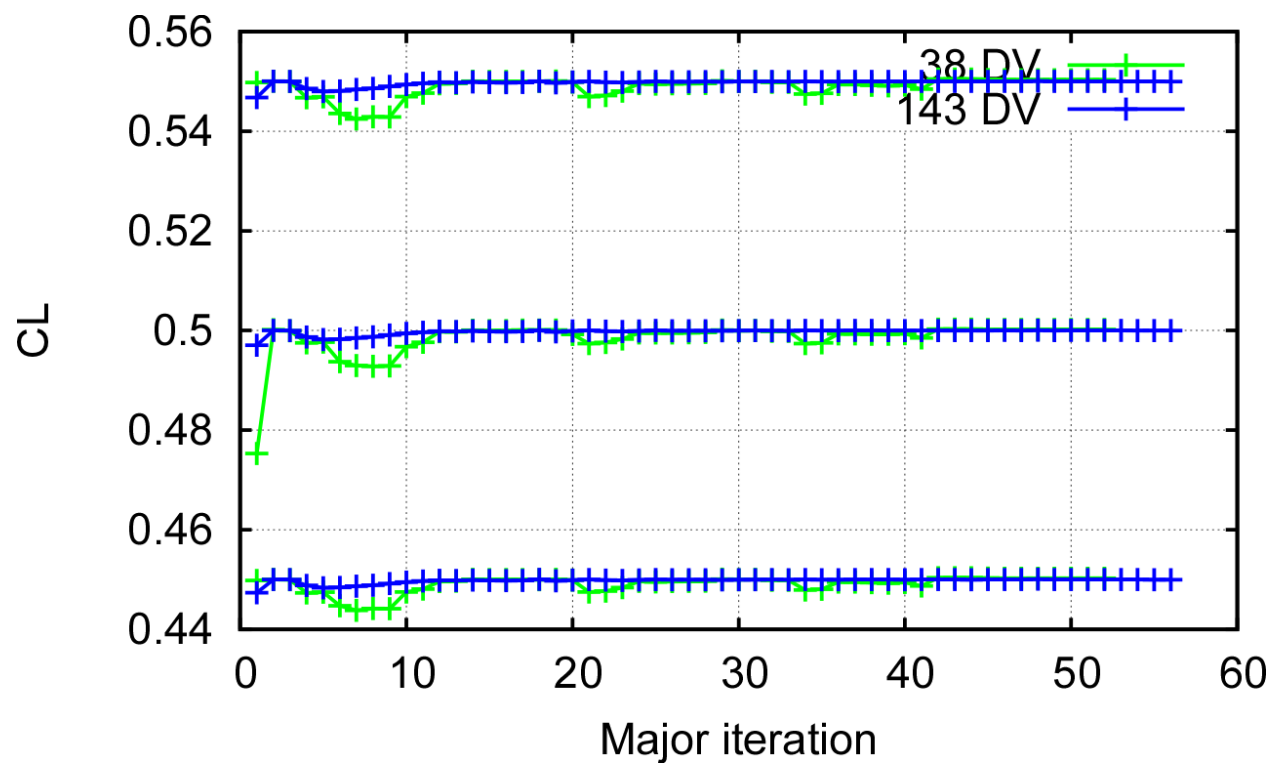
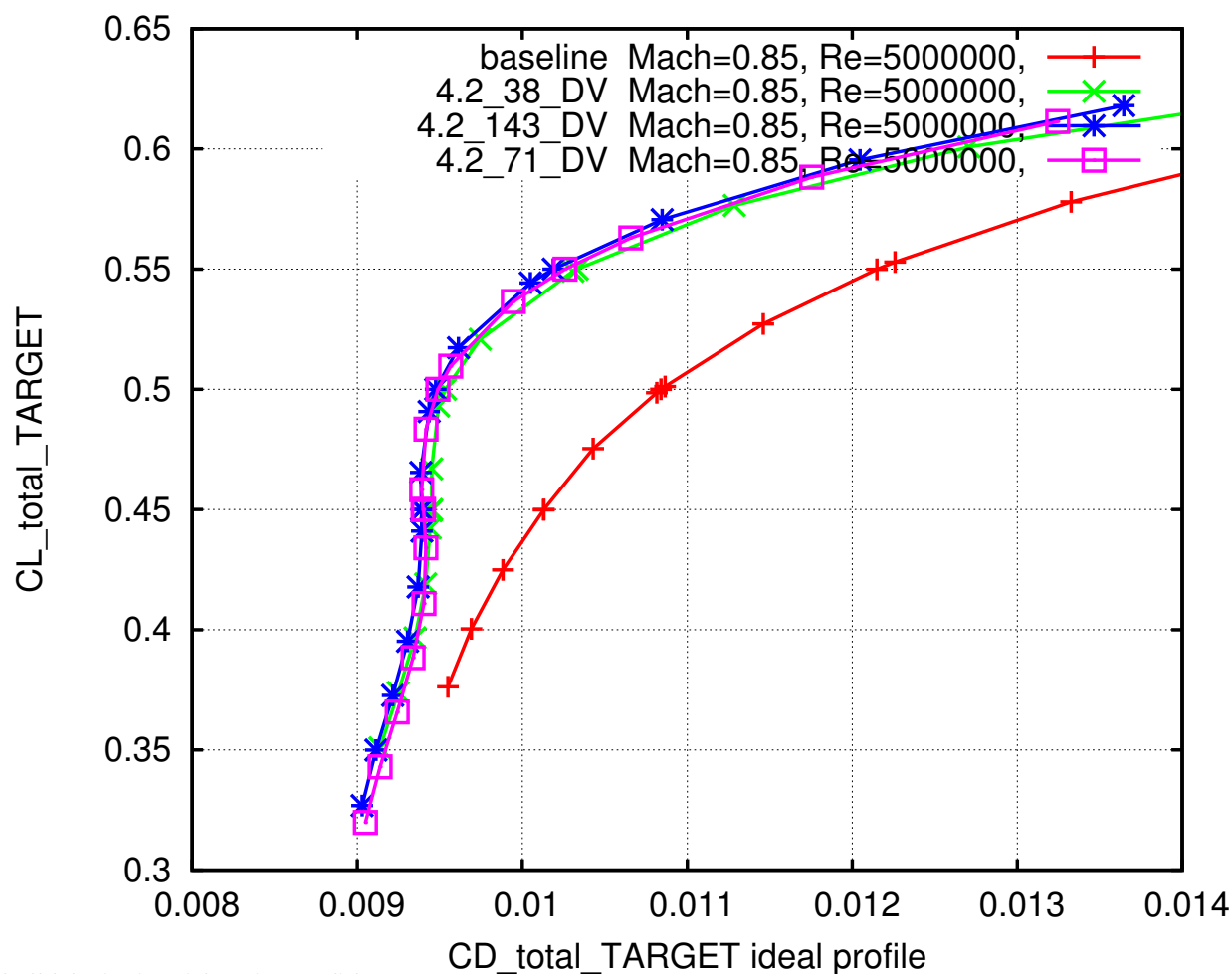
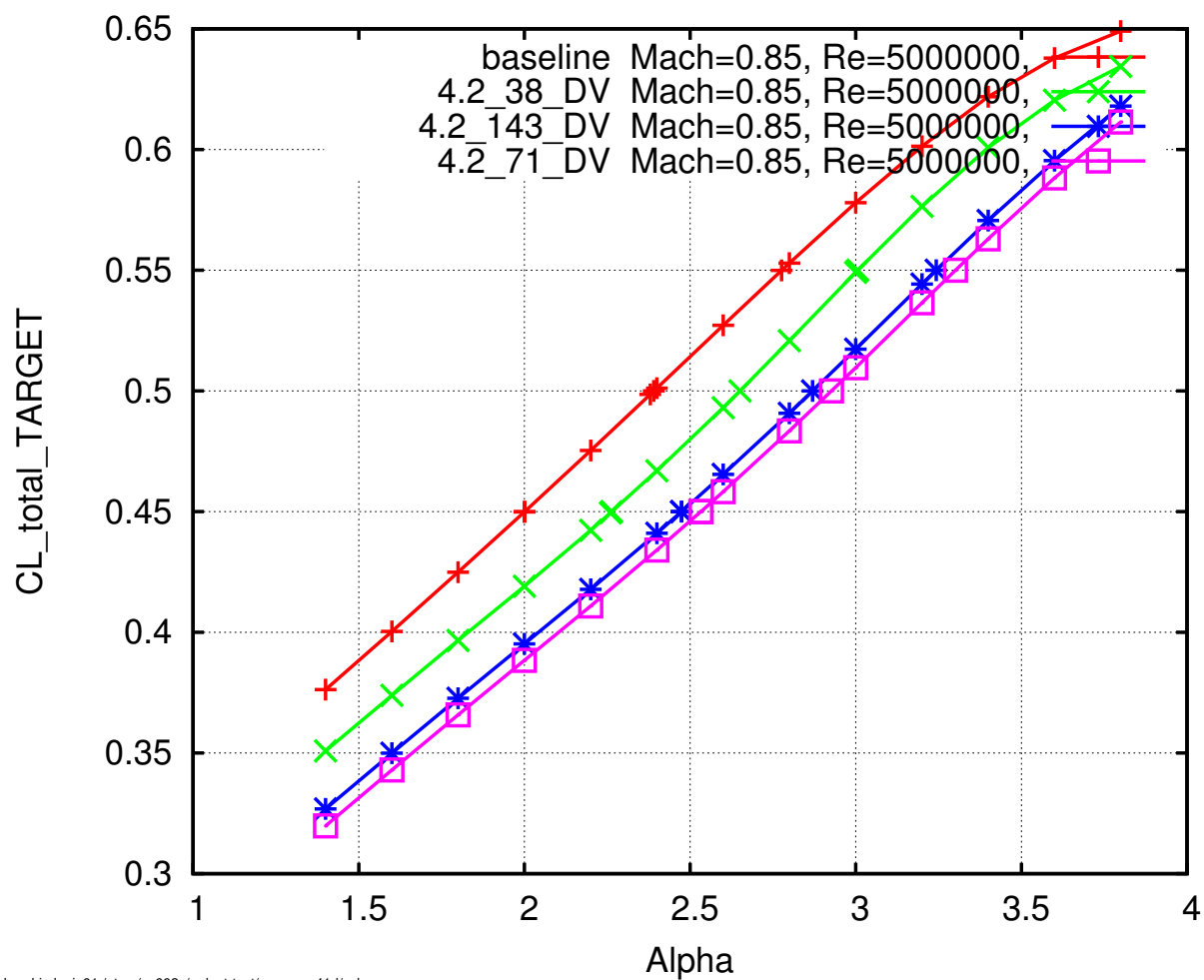


Figure 11. Convergence of the Lift for crm42-overadj38 and crm42-overadj143 Multipoint Designs.



handriv.lrvninf01-intmn/nu6R2n/mrdnt.taet/nrm.caeed41d/erhncsee

Figure 12. Drag Polars for the OVERADJ Multipoint Designs Compared to the Baseline Wing With the Single Cell Trailing Edge Gridding.



handriv.lrvn01-intm/nv6R2n/mfnt.taet/nrm.caeed1d/erhncsee

Figure 13. Lift Curve for the OVERADJ Multipoint Designs Compared to the Baseline Wing With the Single Cell Trailing Edge Gridding.

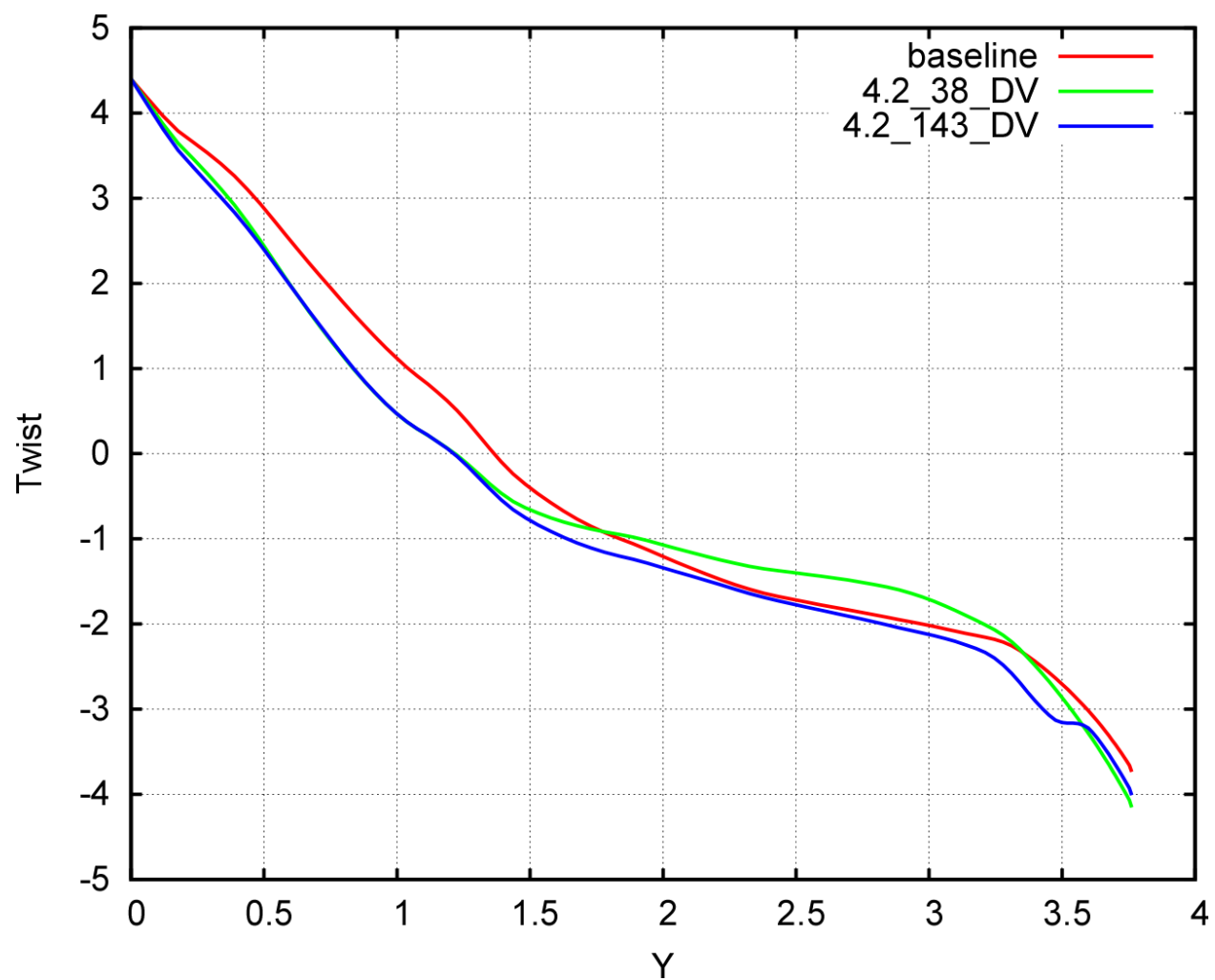


Figure 14. Twist Distributions for crm42-overadj38 and crm42-overadj143 Multipoint Designs Compared to the Baseline Wing.

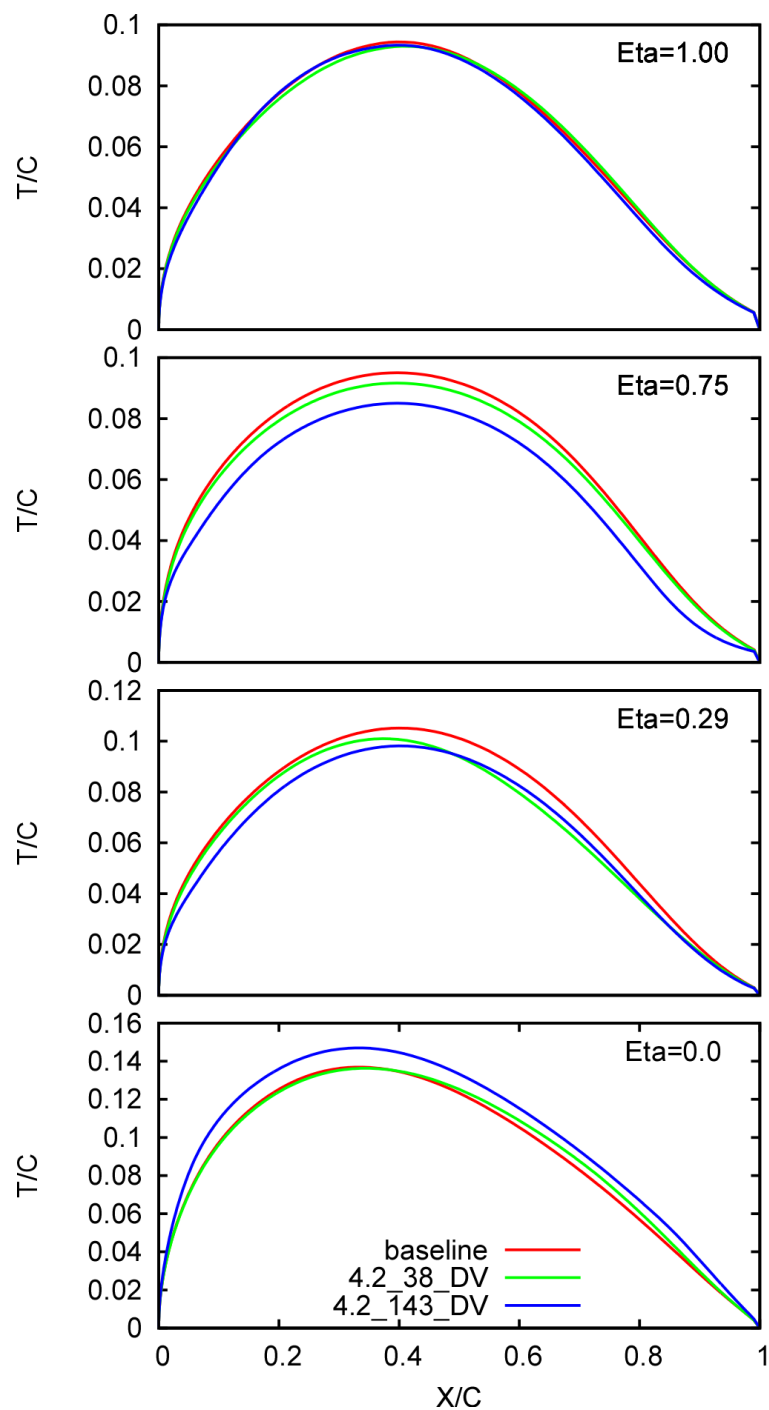


Figure 15. Thickness Distributions for crm42-overadj38 and crm42-overadj143 Multipoint Designs Compared to the Baseline Wing.

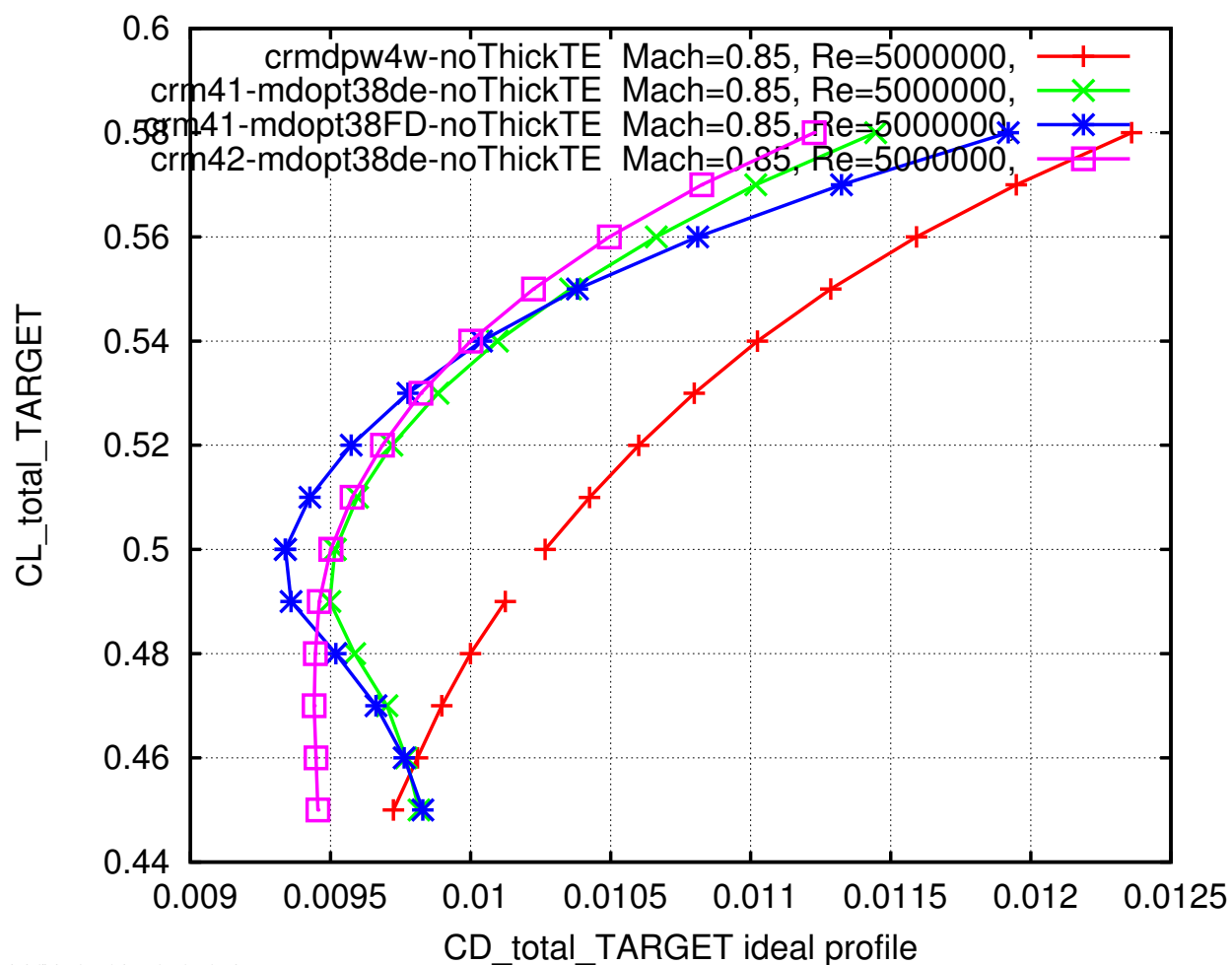


Figure 16. OVERFLOW Drag Polars for crm41-mdopt38de compared to crm41-mdopt38fd and crm42-mdopt38de With Single Grid Cell Trainling Edge Gridding.

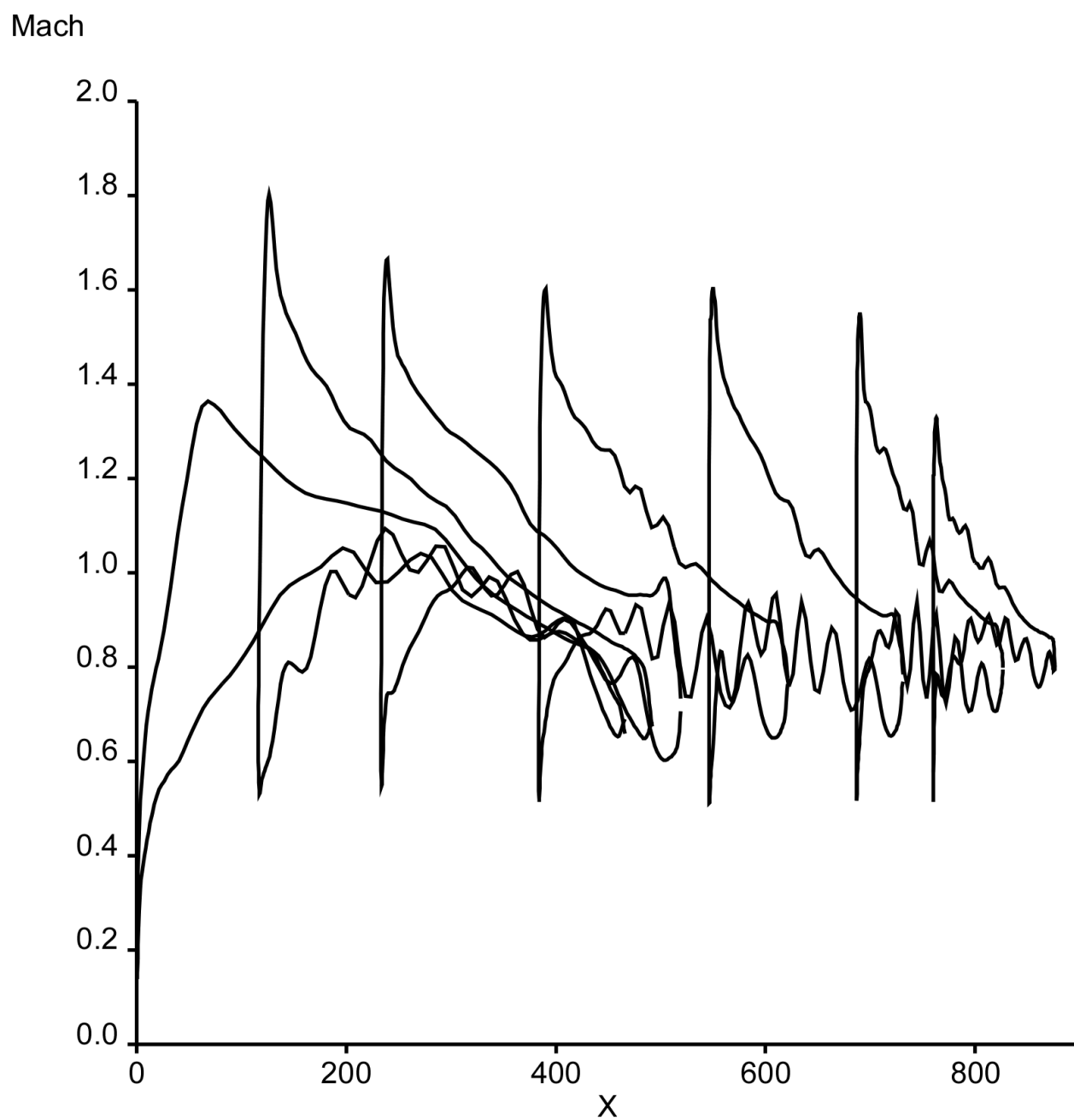
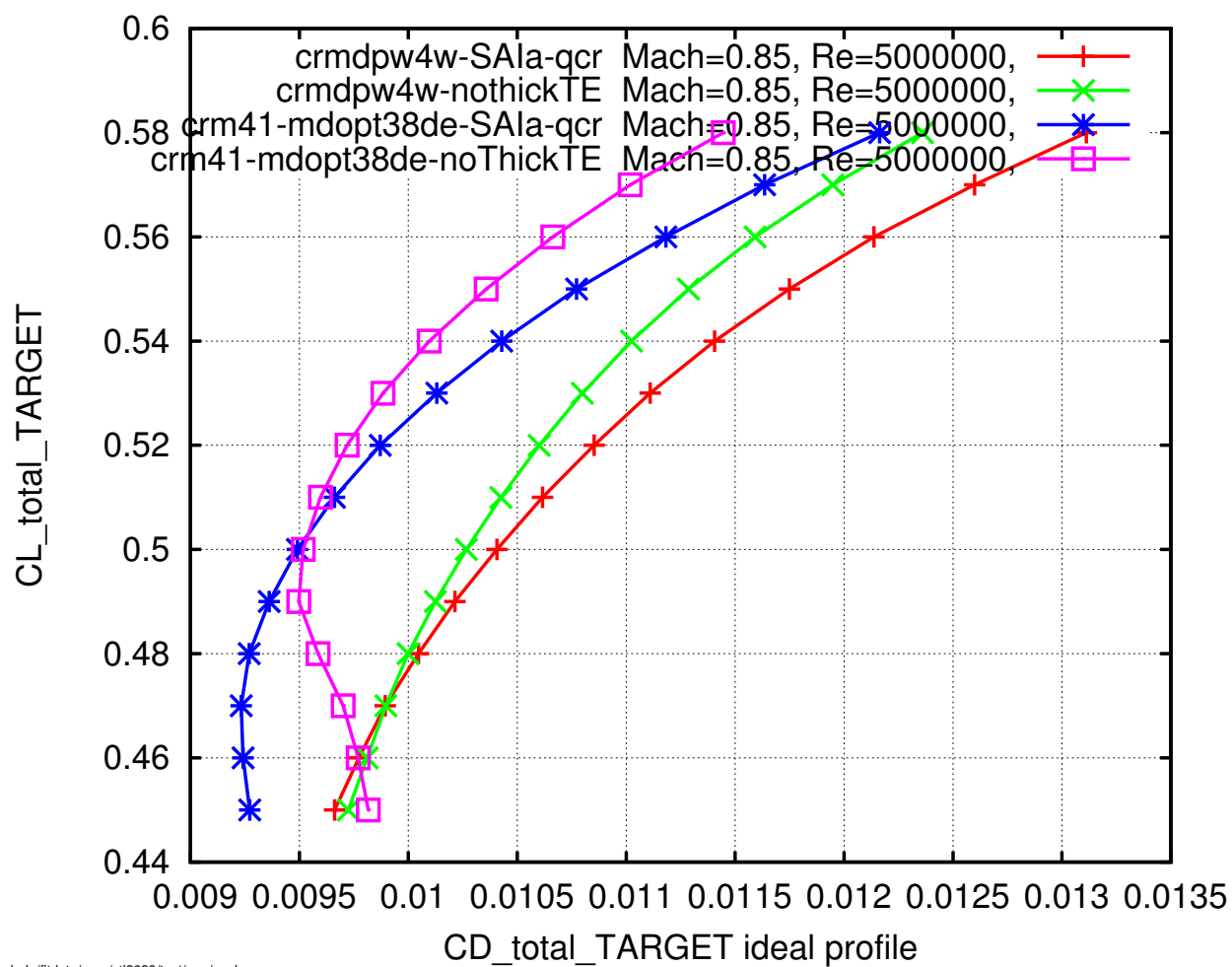
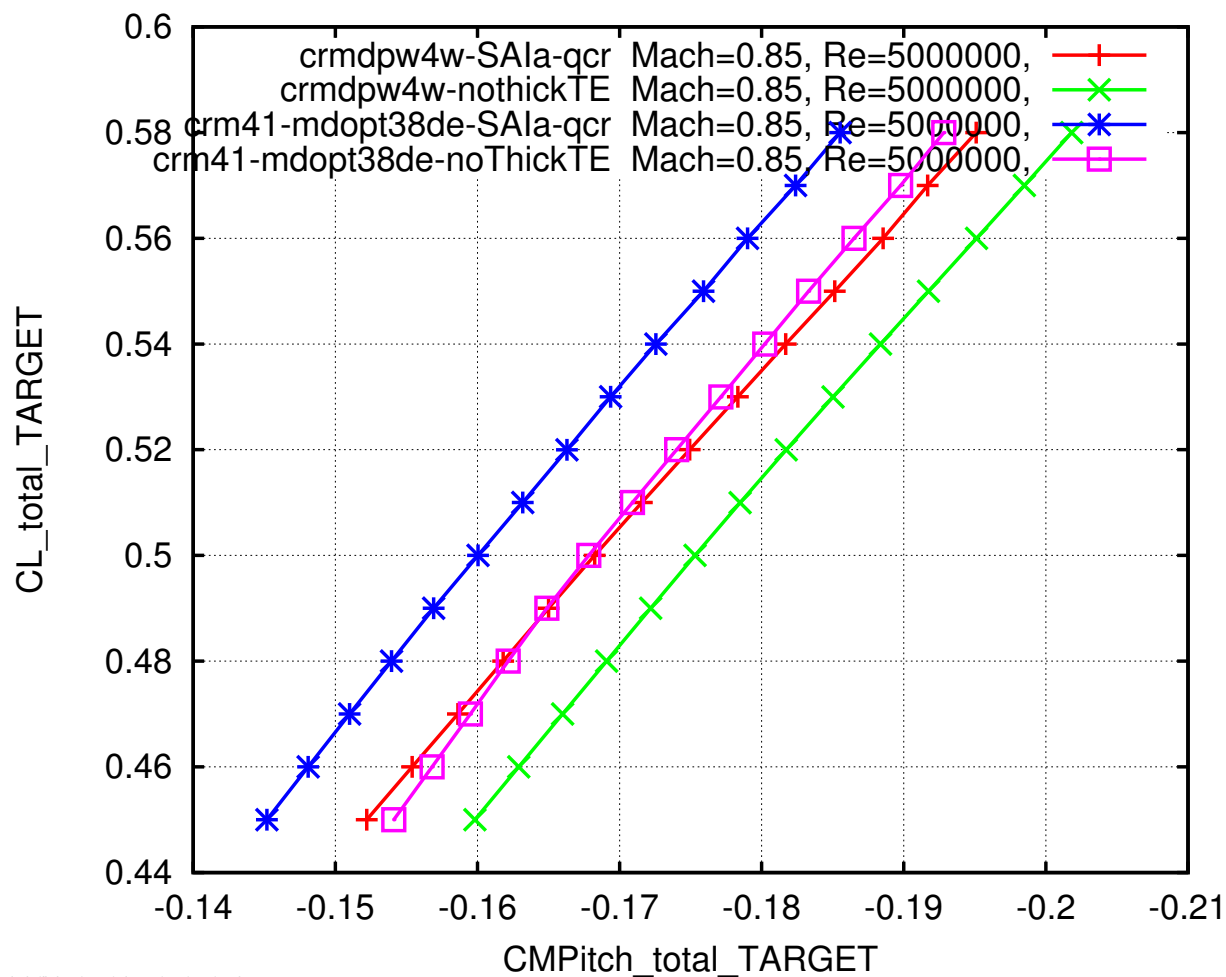


Figure 17. Mach Distribution for crm41-tran736a.



clark:/ftrdata/aern/etl2620/test/crm/eval

Figure 18. Polars for the Baseline Wing and crm41-mdopt38de using Single Cell Trailing Edge Closure and Full Gridding.



ark:/61904/arc:et/2620/taet/crm/aval

Figure 19. Pitching Moments for the Baseline Wing and crm41-mdopt38de using Single Cell Trailing Edge Closure and Full Gridding.

CL

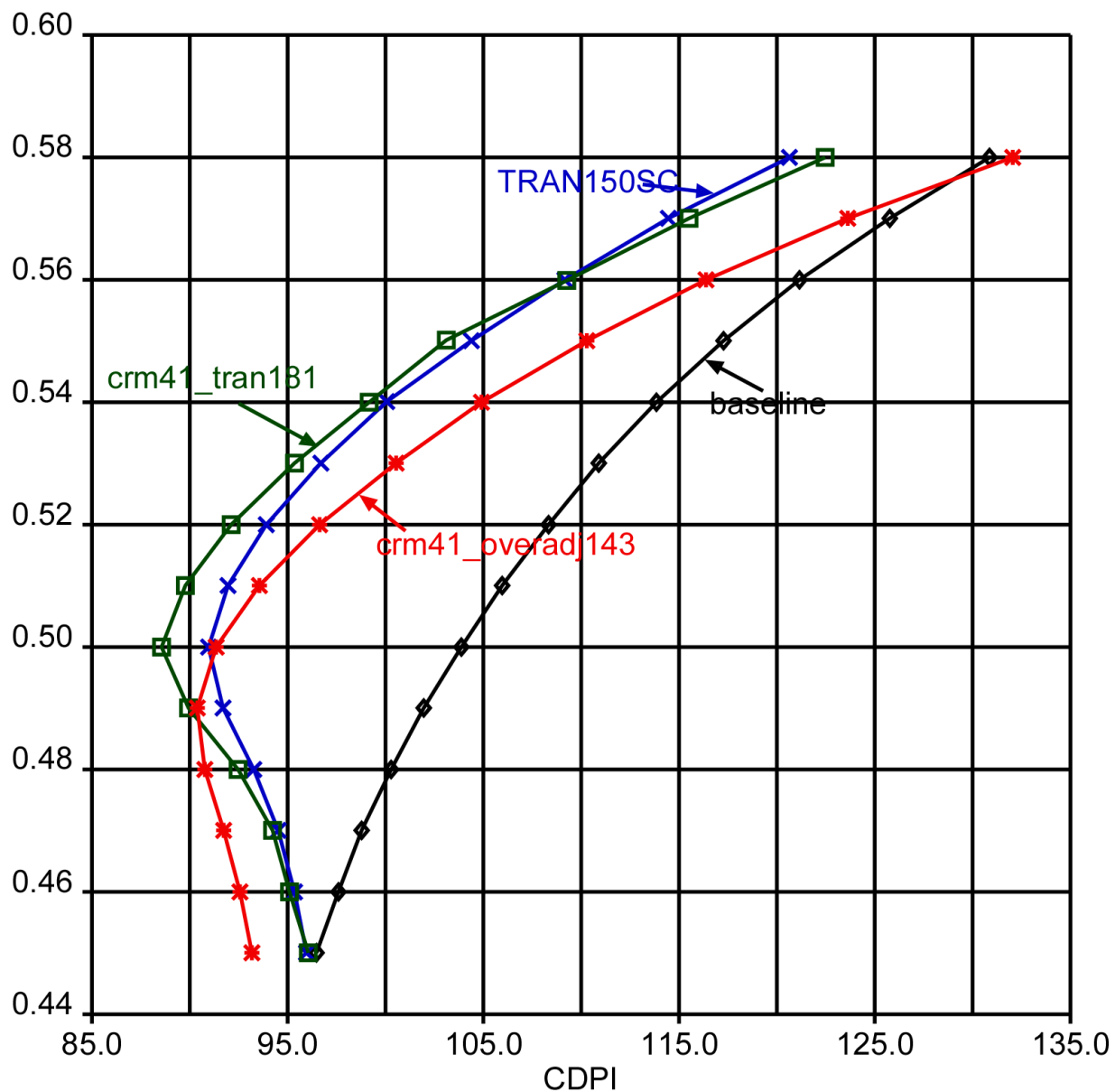


Figure 20. OVERFLOW/SARC/QCR Polars for some of the Single Point Designed Wings.

CL

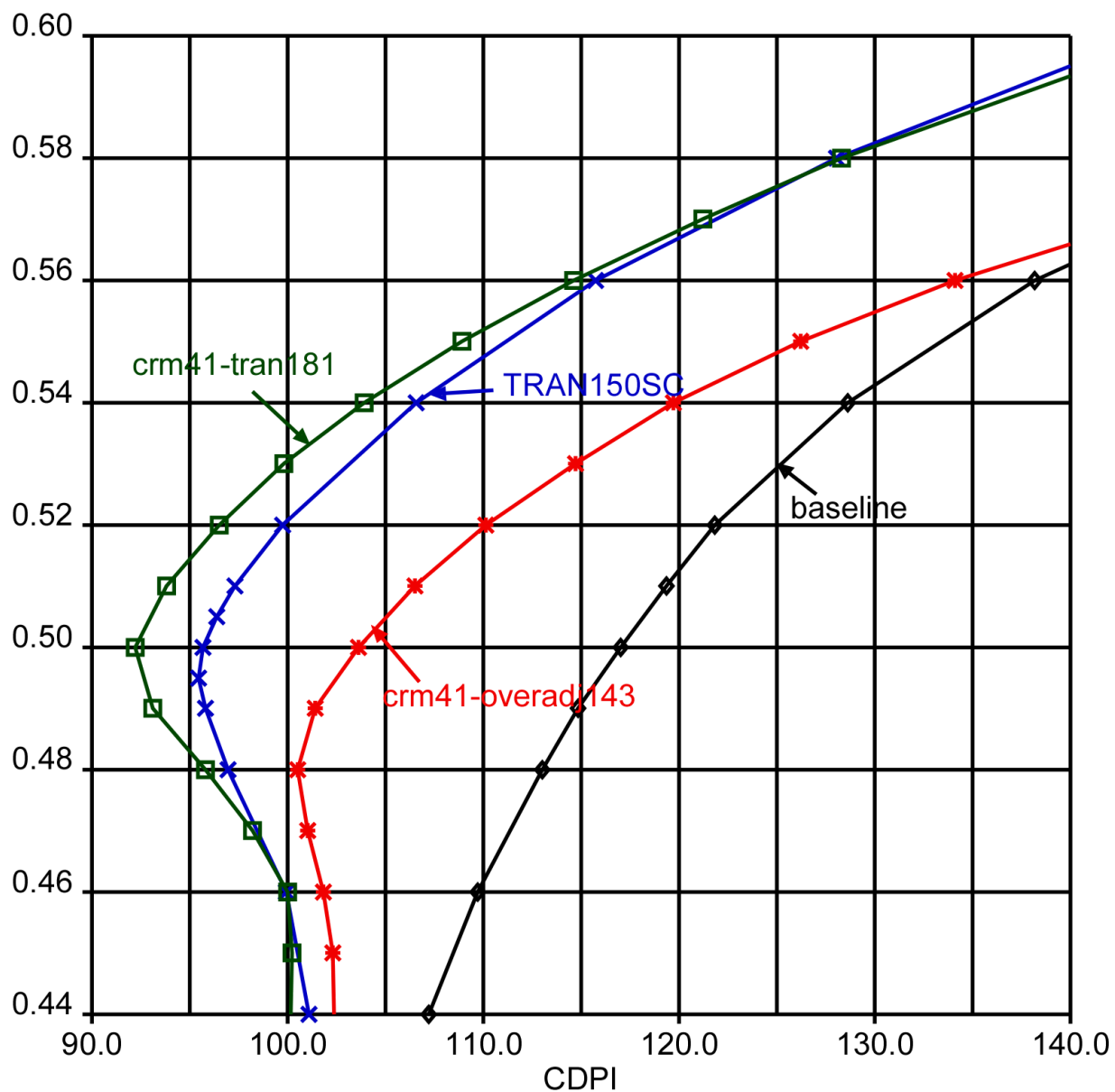


Figure 21. TRANAIR Polars for some of the Single Point Designed Wings.

CL

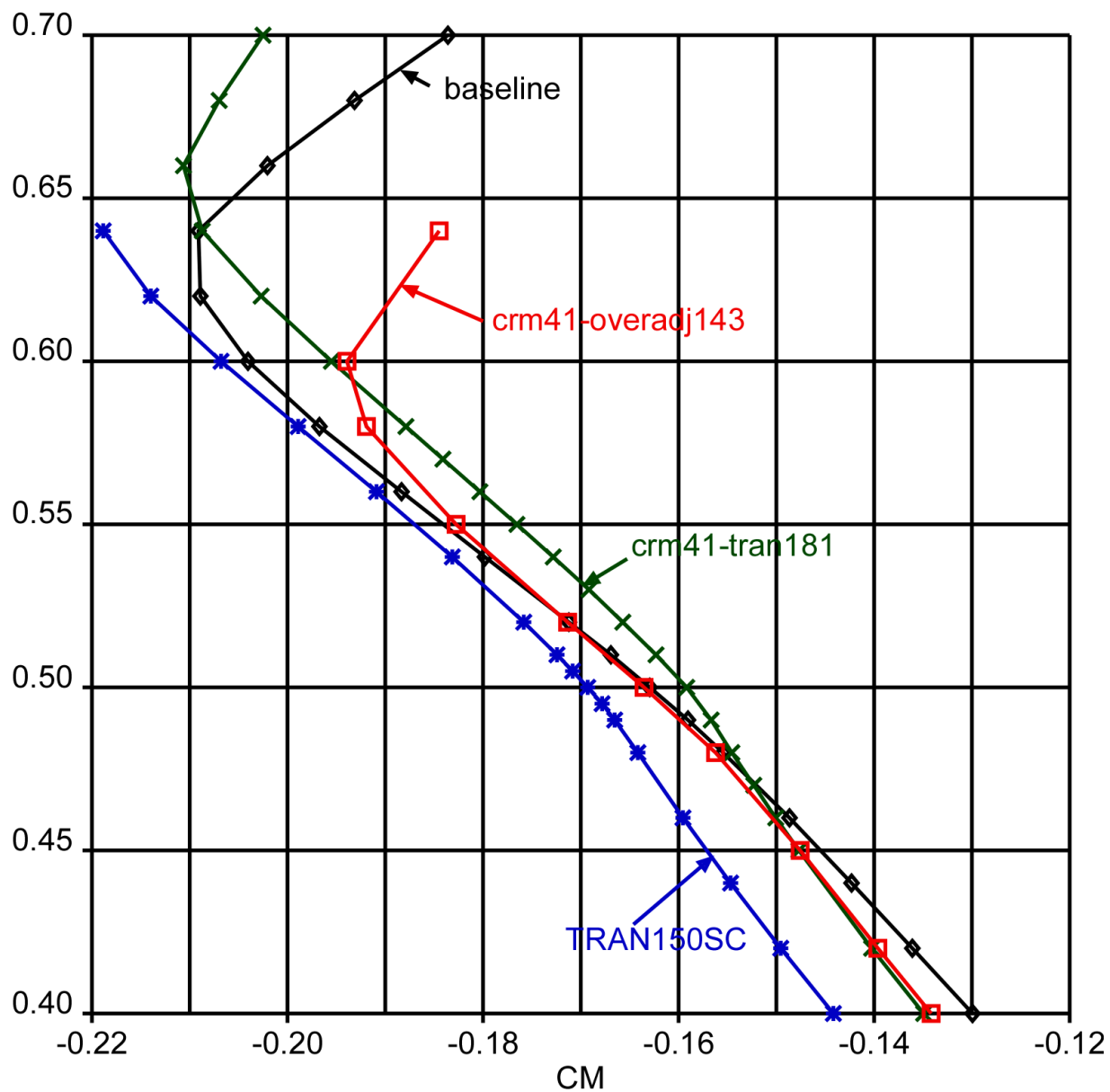


Figure 22. TRANAIR C_M Results For Some of the ADODG CRM Designed Wings.

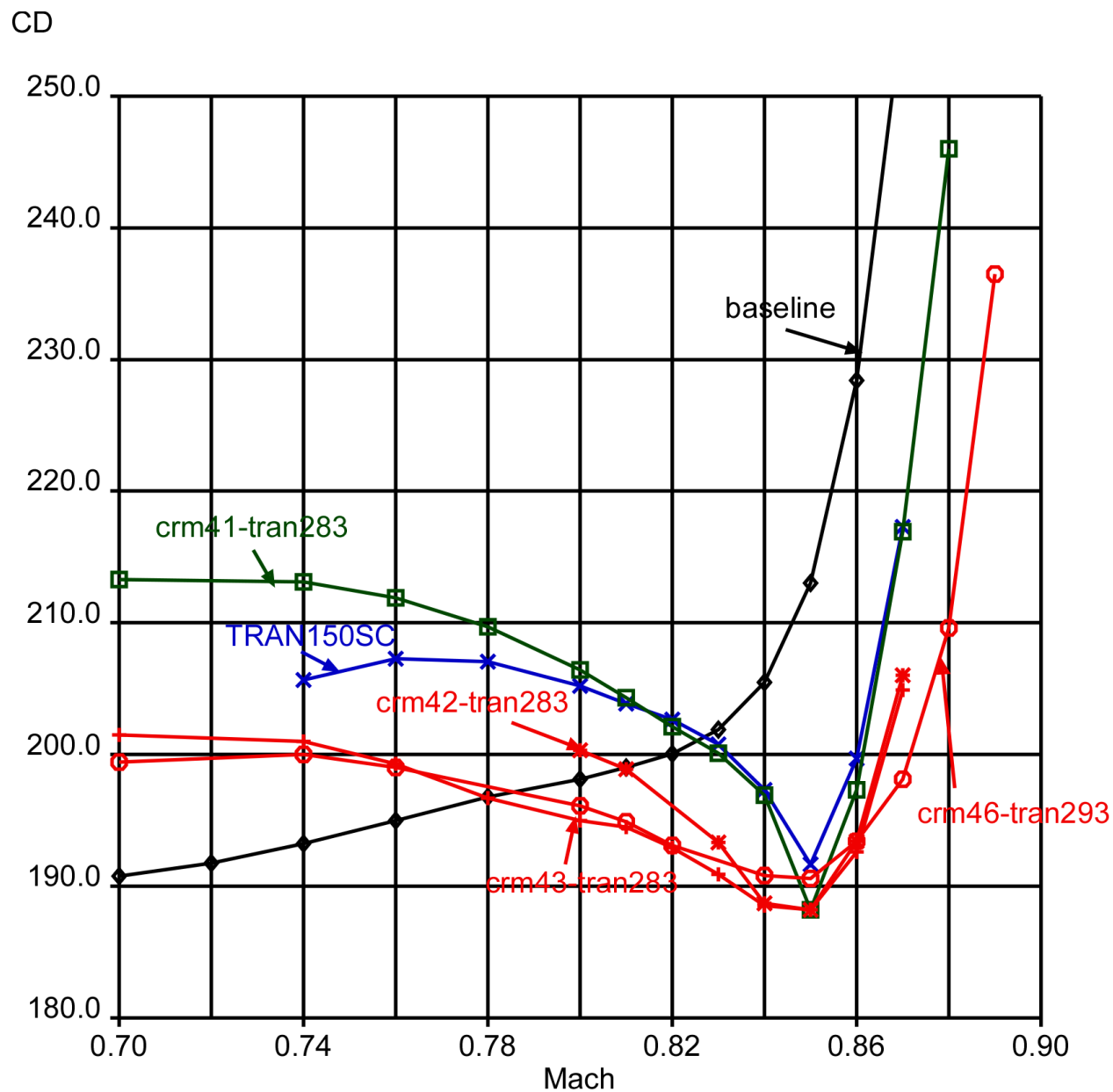


Figure 23. TRANAIR Drag Rise Results For Some of the ADODG CRM Designed Wings.

CL

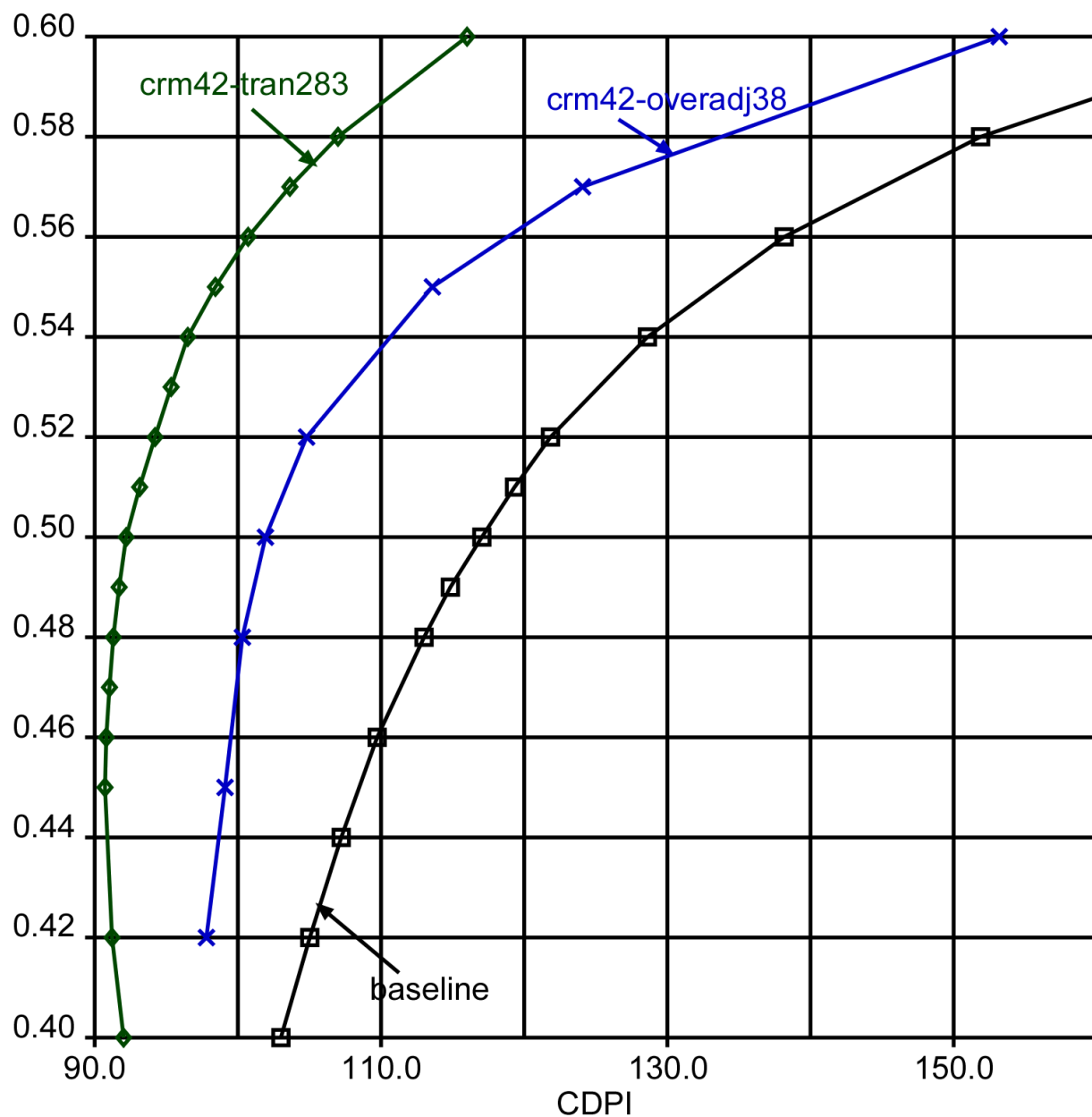


Figure 24. TRANAIR Polars for Some of the crm42 Threepoint Designed Wings.

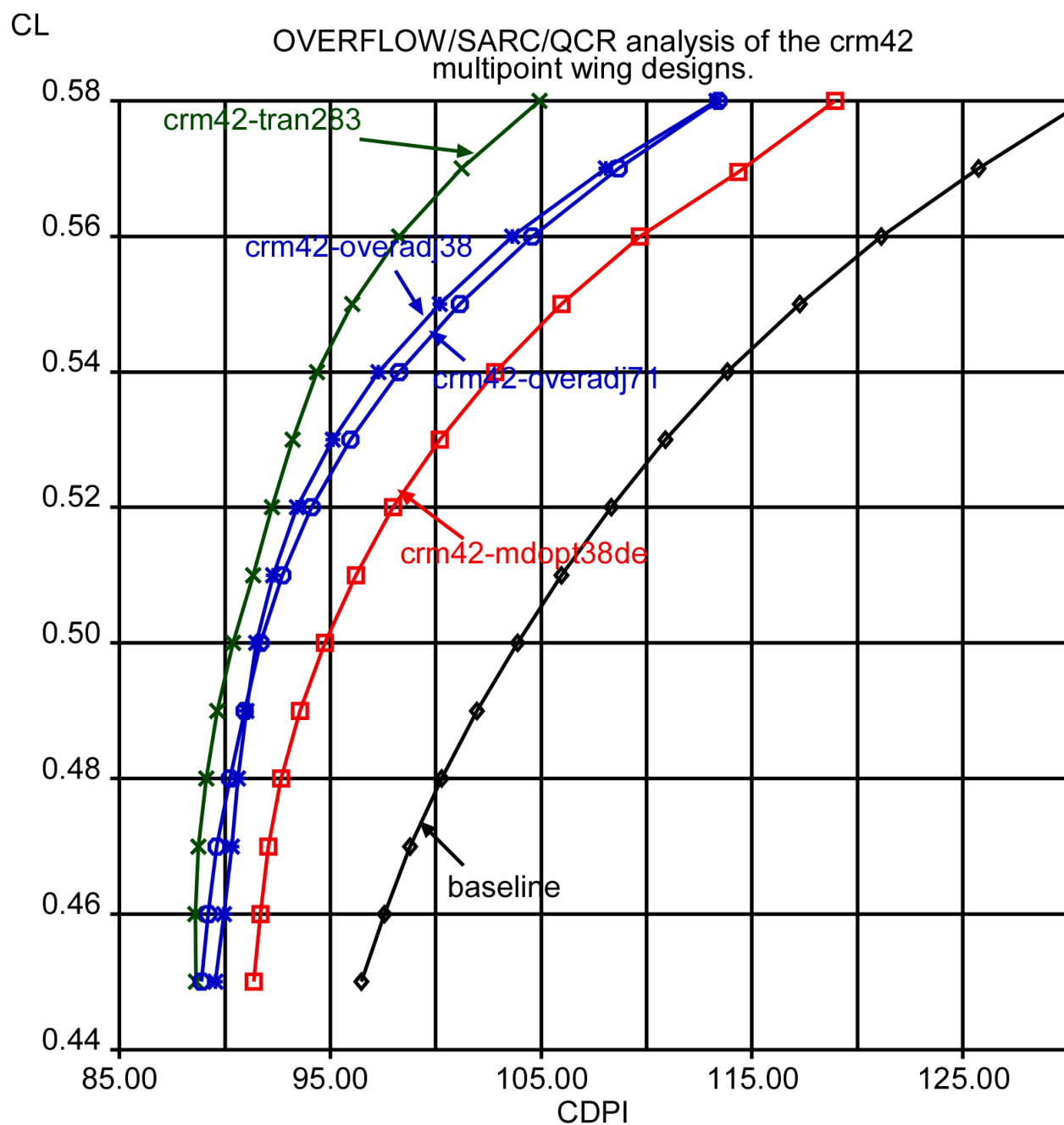


Figure 25. OVERFLOW/SARC/QCR Polars for Some of the crm42 Threepoint Designed Wings.

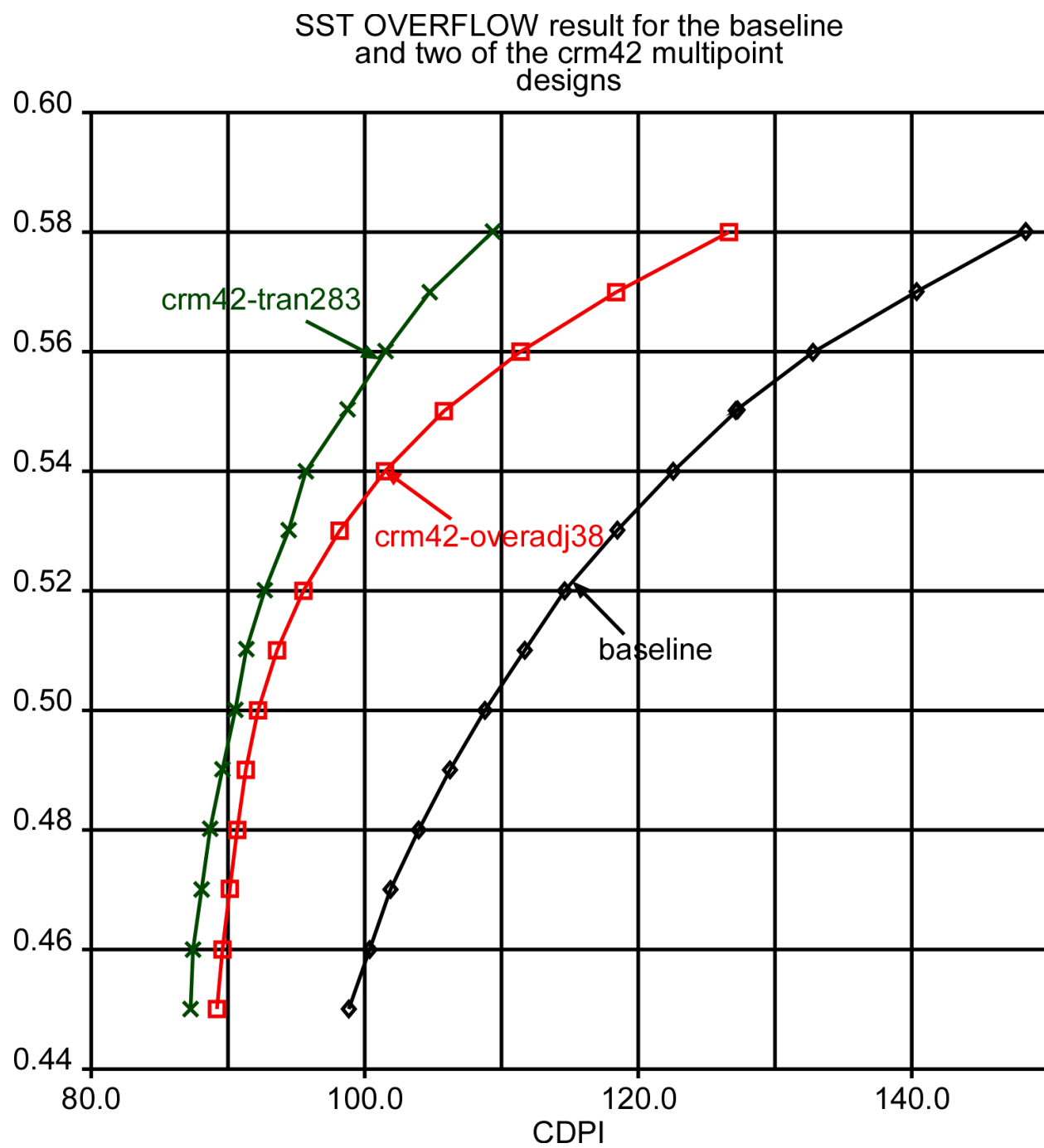


Figure 26. OVERFLOW/SST Polars for Some of the crm42 Threepoint Designed Wings.

Z

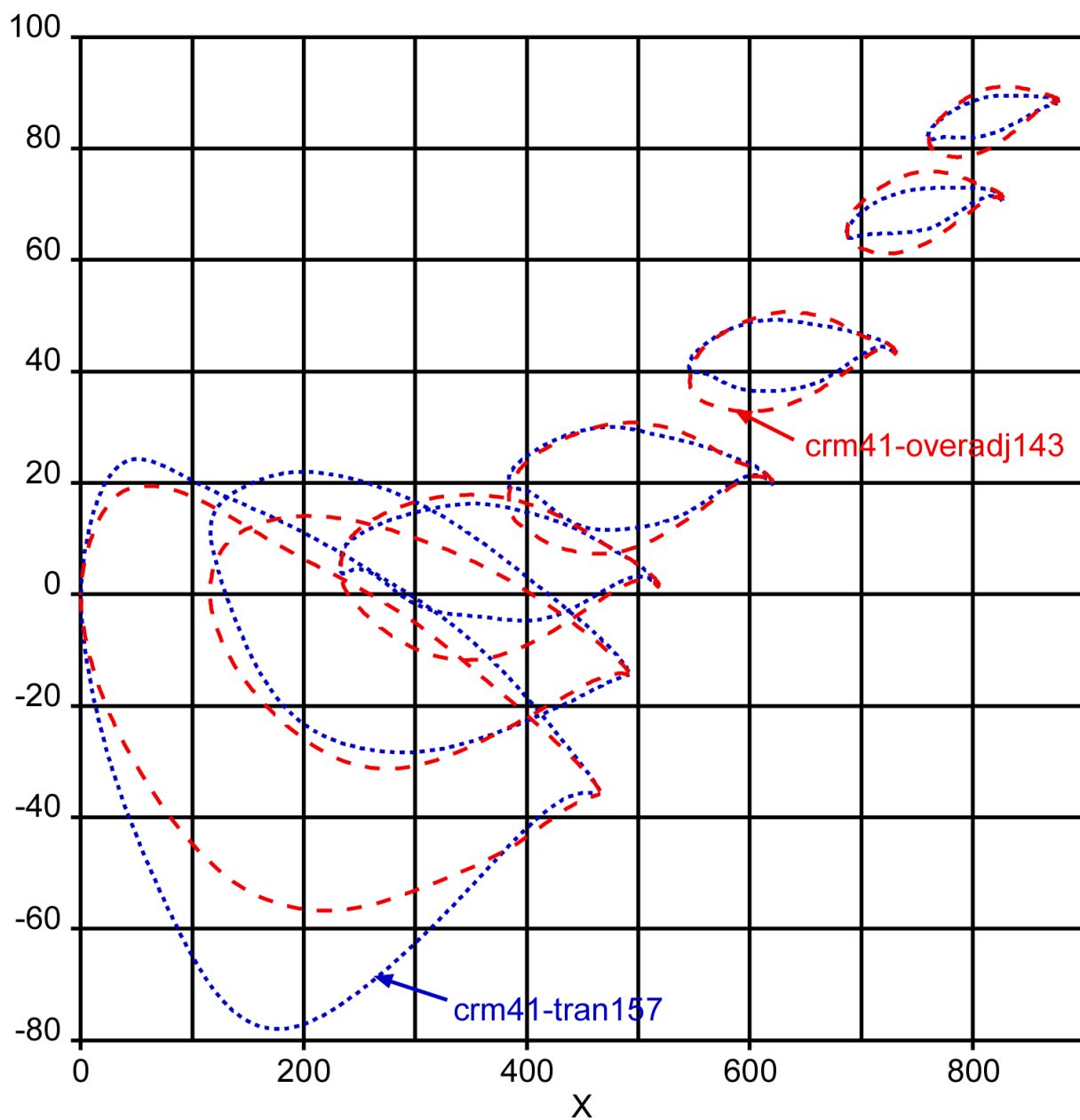


Figure 27. Wing Geometries for crm41-tran157 vs. crm41-overadj143 vs. the ADODG CRM Baseline.

Cp

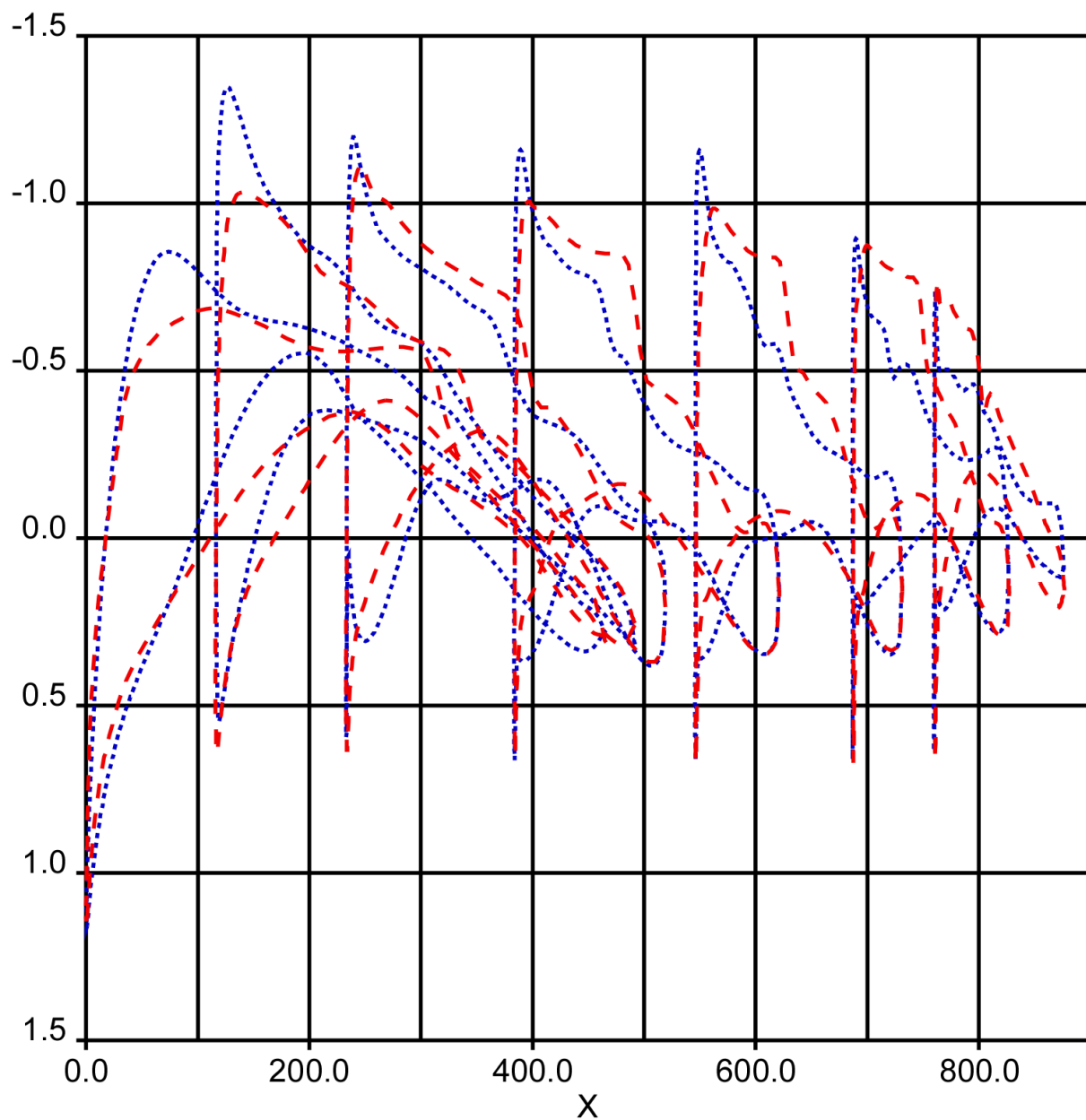


Figure 28. C_p Distributions for crm41-tran157 vs. crm41-overadj143.

CL

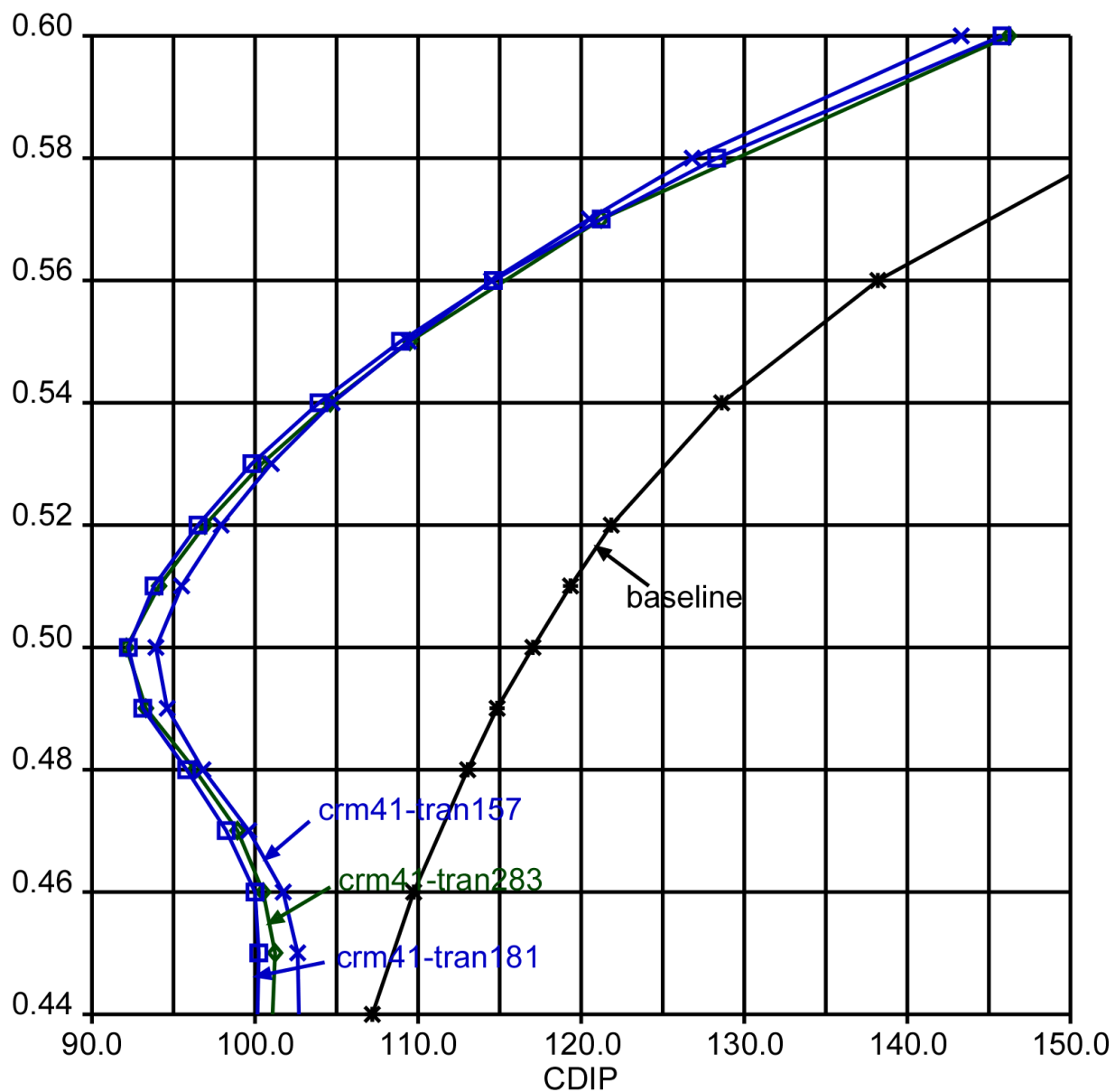


Figure 29. TRANAIR polars for crm41-tran157, crm41-tran181, and crm41-tran283.

CL

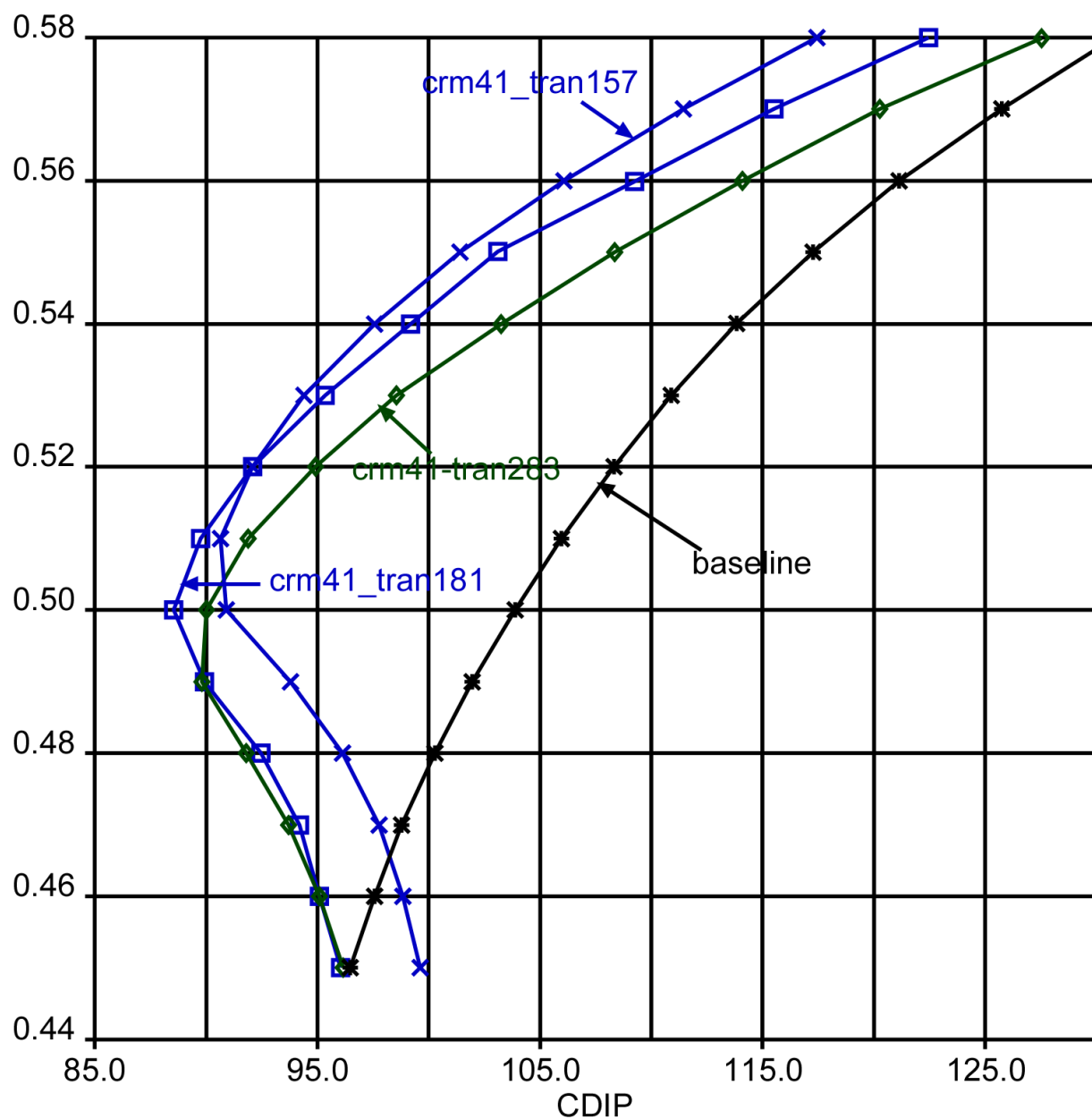


Figure 30. OVERFLOW/SARC/QCR polars for crm41-tran157, crm41-tran181, and crm41-tran283.

CL

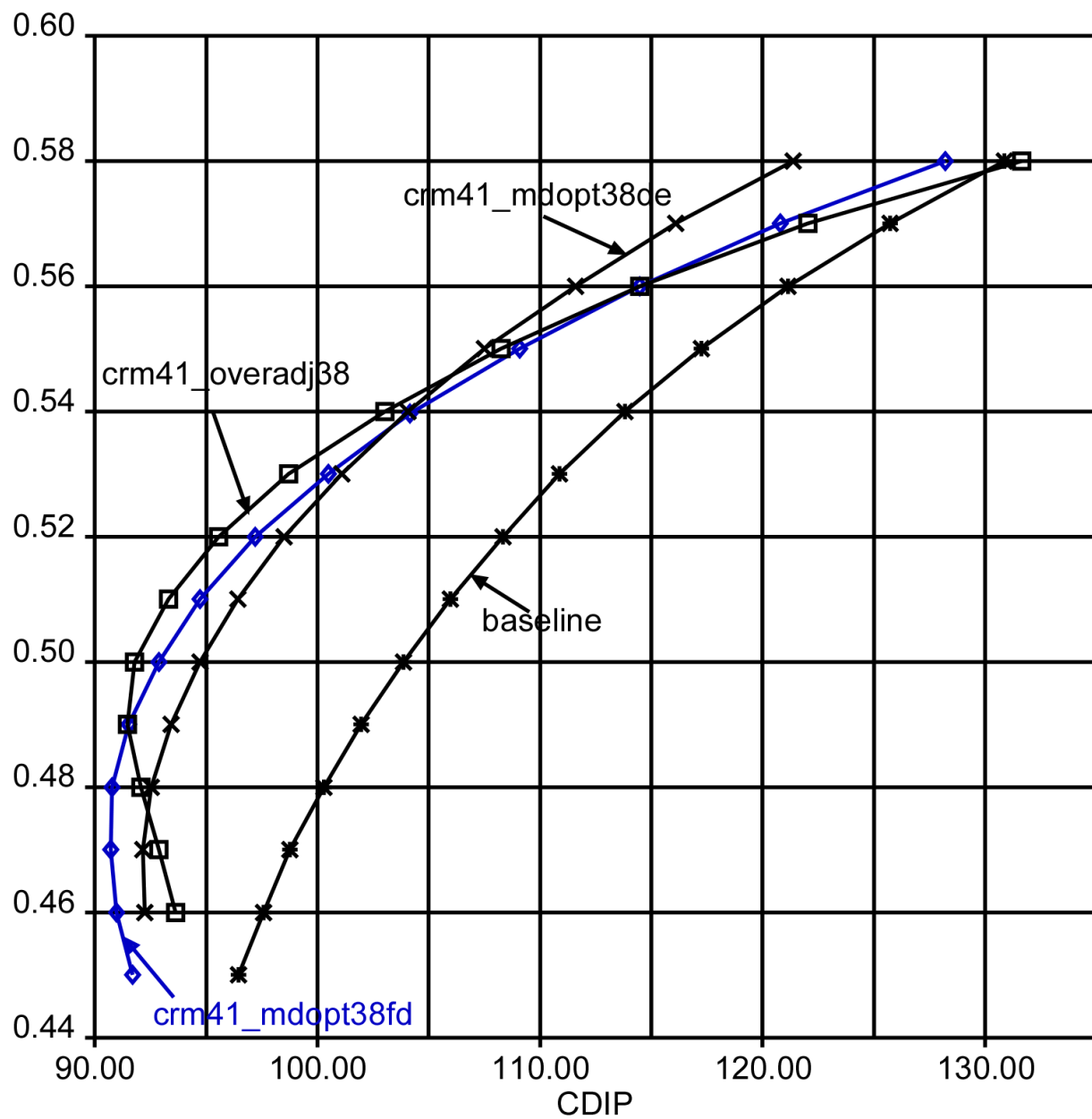


Figure 31. OVERFLOW/SARC/QCR polars for crm41-mdopt38de, crm41-mdopt38fd, and crm41-overadj38.

CL

Polars for the Baseline vs. crm41-tran181
with 3 different modeling options.

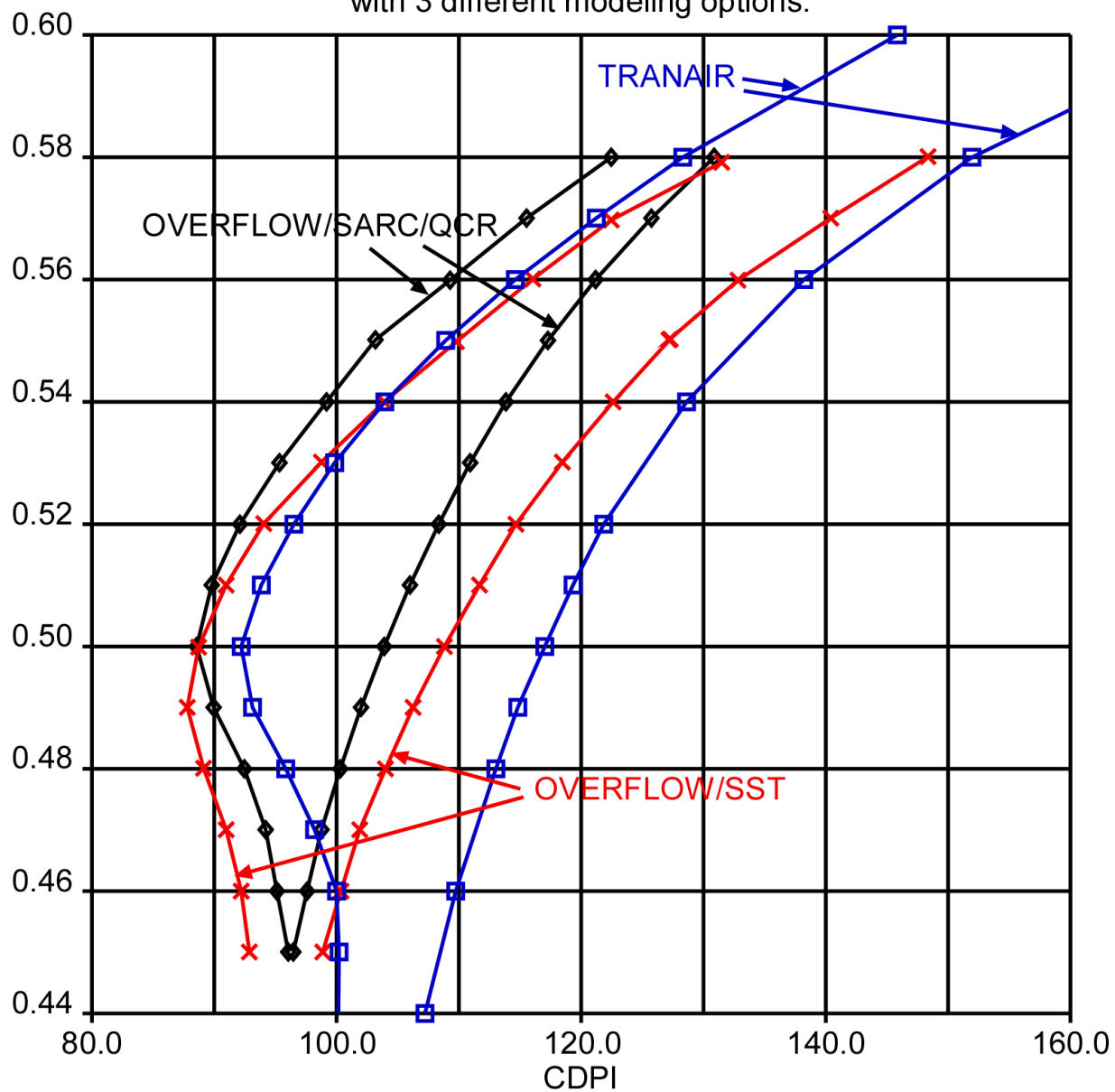


Figure 32. Polars for crm41-tran181 and the Baseline in TRANAIR, OVERFLOW/SARC/QCR and OVERFLOW/SST.